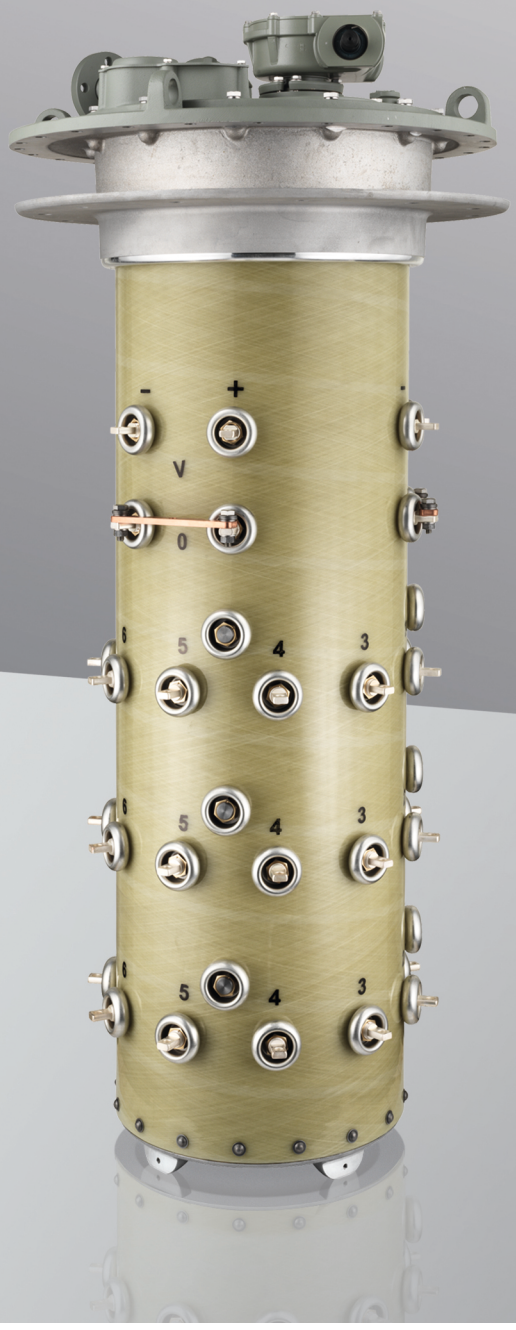
On-Load Tap-Changer VACUTAP® VVS®

Technical Data

5789182/01 EN



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The product may have been altered since this document was published.

We reserve the right to change the technical data, design and scope of supply.

Generally the information provided and agreements made when processing the individual quotations and orders are binding.

The original operating instructions were written in German.

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1 General

1.1 Validity

This file contains the technical data for the VACUTAP® VVS® on-load tap-changer.

This technical file is intended solely for specially trained and authorized personnel.

The associated operating instructions are supplied with the product. These contain descriptions on the safe and proper installation, connection, commissioning and monitoring of the product.

Standards quoted

If standards or guidelines are provided as references without the edition (year of publication) being stated, the version applicable when this document went to print shall apply.

1.2 Manufacturer

The product is manufactured by:

Maschinenfabrik Reinhausen GmbH

Falkensteinstraße 8
93059 Regensburg, Germany
Tel.: (+49) 9 41/40 90-0
Fax: (+49) 9 41/40 90-7001
E-mail: sales@reinhausen.com

Further information on the product and copies of this technical file are available from this address if required.

1.3 Completeness

This technical file is incomplete without the supporting documentation.

1.4 Supporting documents

The following documents also apply in addition to this technical file:

- Technical data and drawings which are provided upon order confirmation and delivery

Also observe generally valid legislation, standards, and guidelines as well as specifications on accident prevention and environmental protection in the respective country of use.

2 Technical data

On-load tap-changer	VVS III 400 Y	VVS III 400 D
Number of phases and application	3, neutral point	3, at any point in the winding
Max. rated through-current Ium (in A)	400	400
Rated short-time current (in kA)	4	4
Rated duration of short-circuits (in s)	3	3
Rated peak withstand current (in kA)	10	10
Maximum rated step voltage U_{in} (in V)	1300	1300
Step capacity (PStN, in kVA)	400	400
Rated frequency (in Hz)	50, 60	50, 60
Operating positions	with change-over selector: max. 19	with change-over selector: max. 19
Highest voltage for equipment U_m (in kV)[1)]	40, 76	40, 76
Oil compartment	Pressure-tight up to 0.3 bar permanent differential pressure (test pressure 0.6 bar), on-load tap-changer head and on-load tap-changer head cover are vacuum-proof	Pressure-tight up to 0.3 bar permanent differential pressure (test pressure 0.6 bar), on-load tap-changer head and on-load tap-changer head cover are vacuum-proof
Oil suction pipe	fitted as standard	fitted as standard
Oil temperatures at which the on-load tap-changer can be operated in the rated load range	-25 °C...+105 °C	-25 °C...+105 °C
Weight		
On-load tap-changer complete, without oil	approx. 220 kg	approx. 240 kg
On-load tap-changer head cover with integrated motor-drive unit without pressure relief device	approx. 35 kg	approx. 35 kg
Displacement (in dm^3)	170	220
Oil fill quantity V_s and minimum useful volume V_2 of oil conservator[2)] **(in dm^3)**	V_s: 140; V_2: 21 see [► 27]	V_s: 190; V_2: 28 see [► 27]
[1)] In accordance with VDE0111 Part 1: Effective value of the conductor-conductor voltage for which equipment is dimensioned in terms of its insulation.		
[2)] Applies to temperature range of -30 °C…+100 °C		

Table 1: Technical data

3 Electrical properties

This chapter contains information about the electrical properties.

3.1 Through-current, step voltage, step capacity

The through-current is the current which flows through the on-load tap-changer/de-energized tap-changer under normal operating conditions. The level of an on-load tap-changer's through-current generally differs over the set voltage range (e.g. while the transformer's rated power remains constant).

Rated through-current I_u

The maximum through-current which a transformer can continuously carry must be used to rate the on-load tap-changer/de-energized tap-changer. This maximum permitted continuous through-current of the transformer is the rated through-current I_u of the on-load tap-changer/de-energized tap-changer.

Step voltage U_{st}

The step voltage is the operating voltage found between adjacent taps. The step voltage may remain constant or change over the entire set range. If the step voltage is variable, the transformer's maximum step voltage U_{st} is used to rate the on-load tap-changer/de-energized tap-changer.

Maximum rated through-current I_{um}

The maximum rated through-current I_{um} is the maximum design-dependent through-current of an on-load tap-changer/de-energized tap-changer to which the current-related type tests relate.

Rated step voltage U_i

The rated step voltage U_i of an on-load tap-changer is the highest step voltage permissible for one particular value of the rated through-current I_u. Together with a rated through-current, this is known as the associated rated step voltage.

Maximum rated step voltage U_{im}

The maximum rated step voltage U_{im} is the maximum permissible design-dependent step voltage of an on-load tap-changer/de-energized tap-changer.

Transition resistors

The diverter switch's transition resistors are configured depending on the levels of maximum step voltage U_{st} and the rated through-current I_u of the transformer for which the on-load tap-changer is intended.

Since the permissible rated through-current I_u and permissible step voltage U_{st} depend on value of the transition resistors, the rated levels relate to the relevant application.

If an on-load tap-changer is operated with step voltage and through-current values other than those stated in the order, Maschinenfabrik Reinhausen GmbH (MR) must check whether this is possible. For example, if the transformer power is increased by means of improved cooling or the on-load tap-changer is used in a different transformer, the transition resistors may have to be adapted.

This also applies if the desired new rated values I_u and U_{st} are below the original values. The transition resistor configuration affects both the switching capacity loading of the contacts and the consistent contact wear.

Rated step capacity P_{StN}

The rated step capacity P_{StN} is the product of the rated through-current I_u and associated rated step voltage U_i:

$$P_{StN} = I_u \times U_i$$

The following diagram shows the typical load limits. This shows that the permissible operating range is limited by the maximum rated step voltage U_{im} and maximum rated through-current I_{um}.

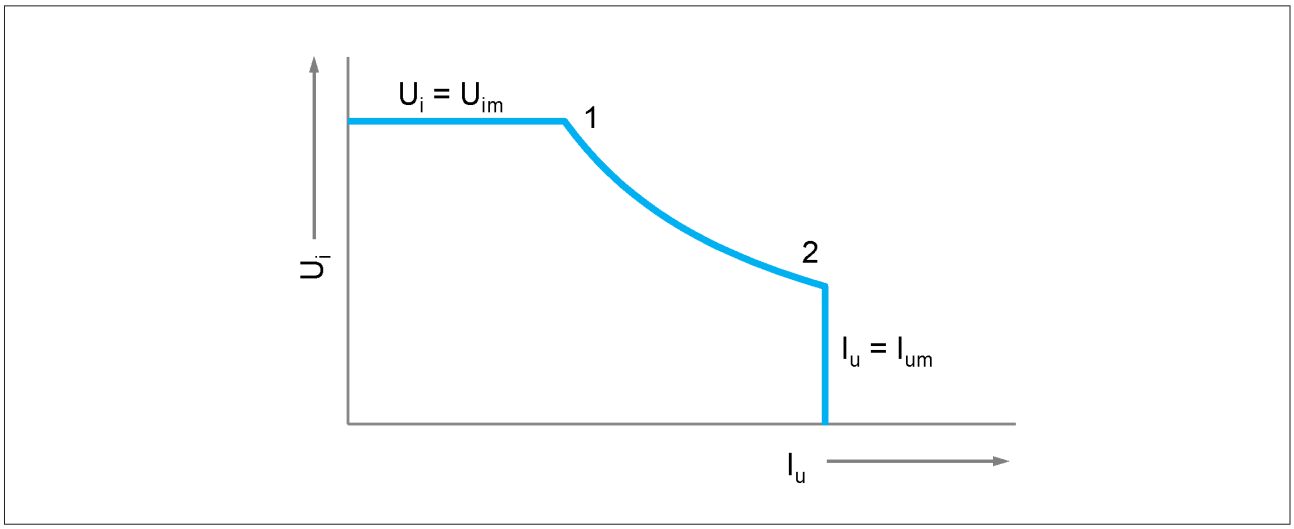


Figure 1: Rated step capacity diagram

1	Upper corner
2	Lower corner

The curve points between corners 1 and 2 are only provided by the permissible rated switching capacity. The permissible rated switching capacity between corners 1 and 2 corresponds to linked pairs of values for I_u and U_i and may be constant or variable.

Limit step capacity and limit switching capacity

The limit step capacity is the largest step capacity which can be switched with certainty. In its standard model, each MR on-load tap-changer can switch at least twice the rated through-current I_u at the step voltage U_{st} for which the on-load tap-changer was configured. This limit switching capacity is proven by type testing in accordance with IEC 60214. Appropriate measures must be taken to prevent tap changes with currents of more than twice the rated through-current I_u.

3.2 Step capacity diagram

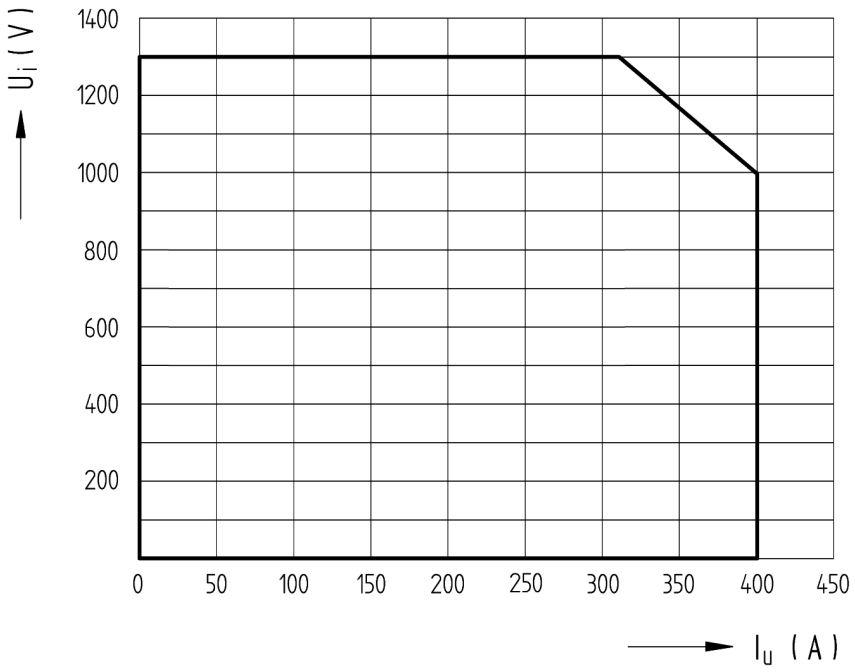


Figure 2: Step capacities (rated step voltage U_i, rated through current I_u)

3.3 Insulation

The stated rated withstand voltages for the insulation arrangement apply to new, perfectly dried insulation in prepared transformer oil (at an ambient temperature of at least 10 °C).

The following details are needed to select an on-load tap-changer:

- Maximum mains-frequency operating voltages
- Test AC voltages during transformer test
- Impulse voltages during transformer test (lightning surge, switching surge, wave isolated in back and wave isolated in front)

The transformer manufacturer is responsible for the correct choice of rated withstand voltages depending on insulation coordination at the operating site. The necessary rated withstand voltages should be observed for the various insulation distances:

- Insulation to ground
- With multi-phase types: Insulation between phases
- Insulation between a phase's contacts

The details required depend on the type of regulation (e.g. basic connection of tapped winding with on-load tap-changers) and the tap-changer type.

3.4 Permitted voltage stresses

This section describes the permitted voltage stresses on the on-load tap-changer.

When selecting the on-load tap-changer, you must check that the highest stresses do not exceed the related rated withstand voltages at the insulation distances.

3.4.1 VACUTAP® VVS-Y insulation distances

Rated voltages of the internal insulation with reversing change-over selector

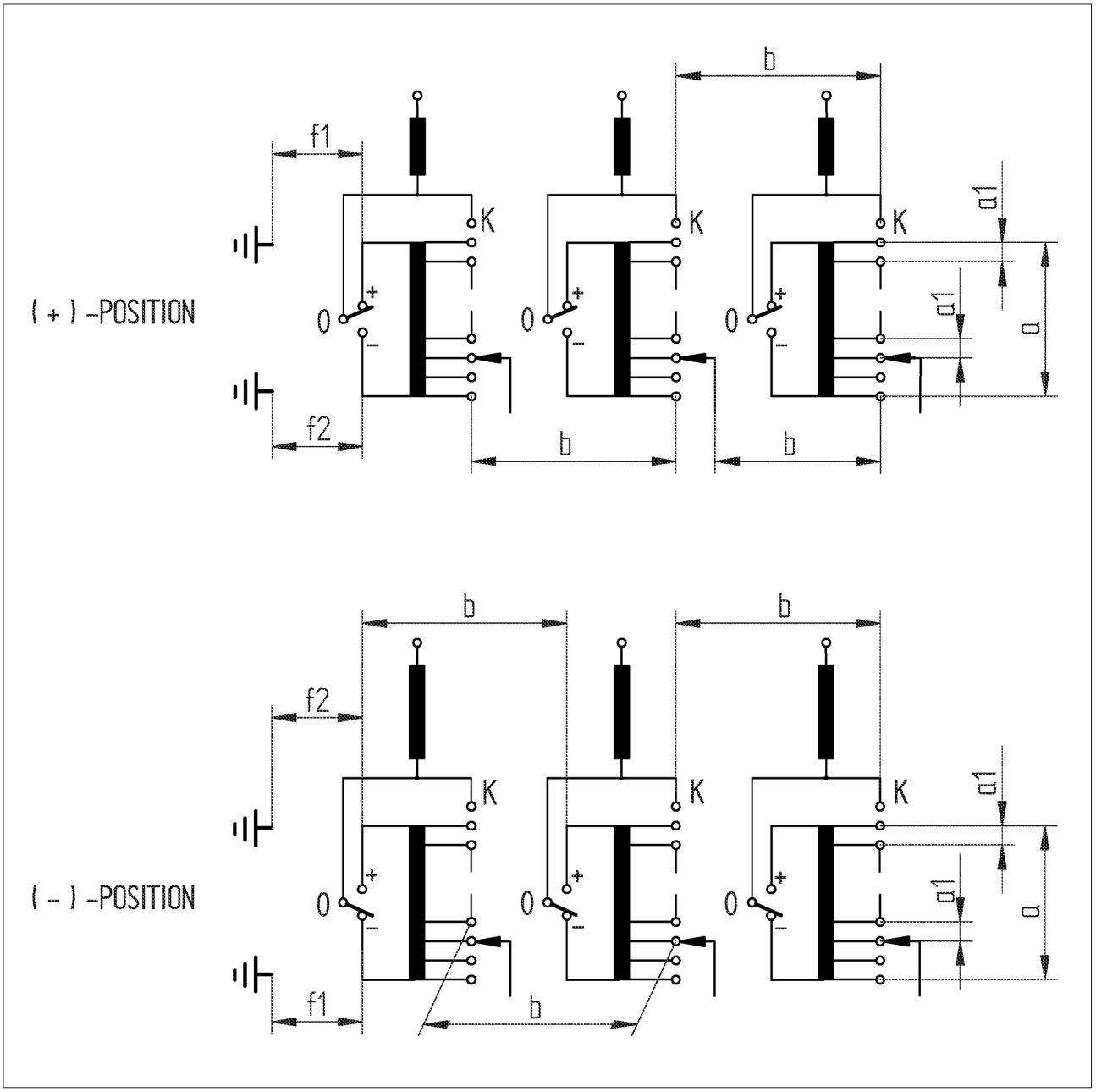


Figure 3: Rated voltages of the internal insulation with reversing change-over selector

Rated voltages of the internal insulation with coarse change-over selector

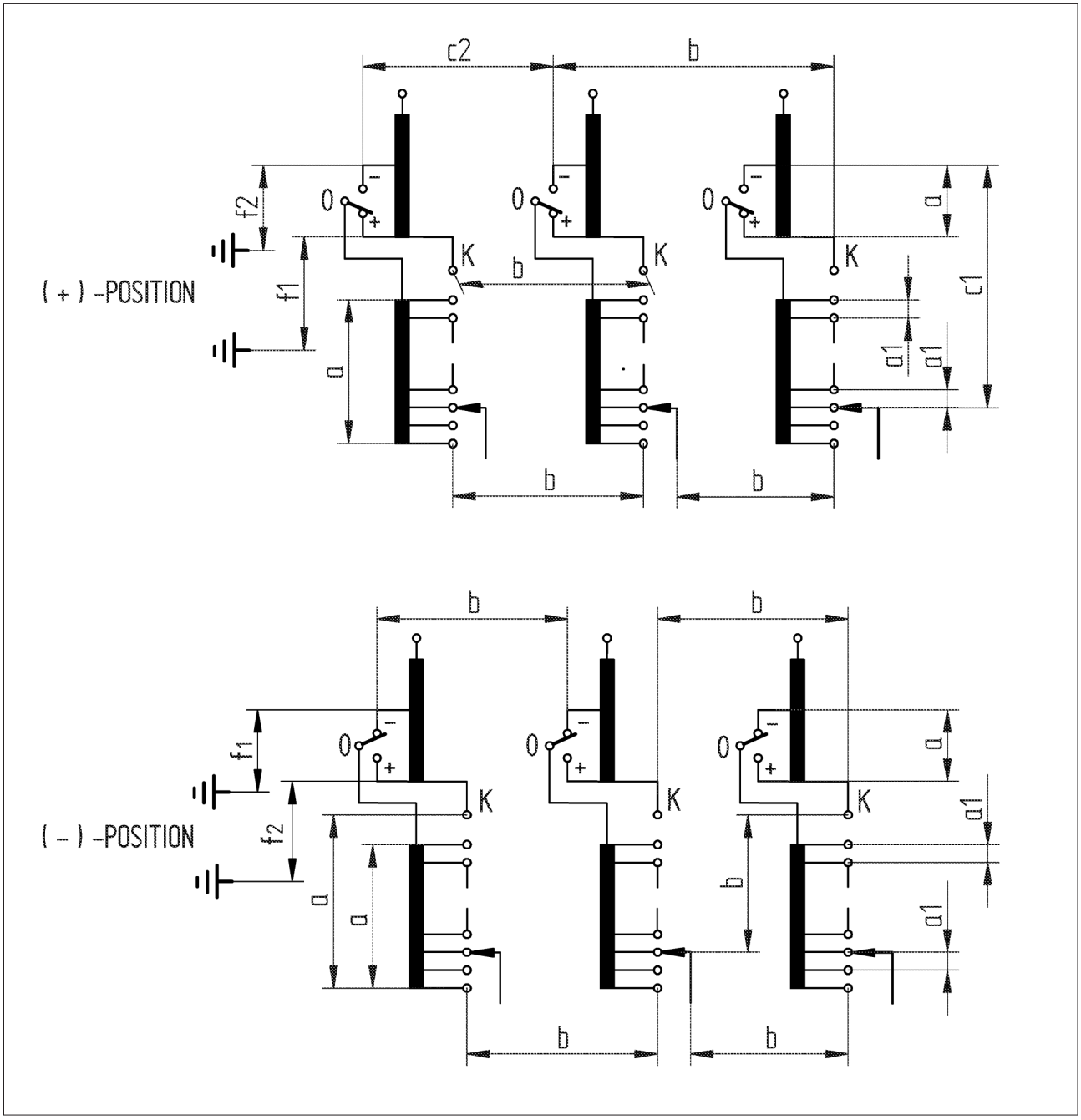


Figure 4: Rated voltages of the internal insulation with coarse change-over selector

a1	= between tap selector contacts of the winding of one tap position (connected or not connected)

a	= between start and end of a tap winding and, in a version with coarse winding, also between start and end of a coarse winding Note for coarse change-over selector connection in (-) position of the change-over selector: When stressed with impulse voltage, note the permissible withstand voltage "a" between the end of a coarse winding connected with the K tap selector contact and the tap selector contact at the end of the tap winding of the same phase.
b	= between the fine tap selector contacts of different phases and between change-over selector contacts of different phases, which are connected with the start/end of a tap winding or with a tap selector contact.
f	= to ground. The following applies to designs with change-over selector:
f_1	= Between selected change-over selector contacts and ground.
f_2	= Between unselected change-over selector contacts and ground.
Additionally for coarse tap selector connection in (+) position of the change-over selector:	
c1	= Between one (-) change-over selector contact and the take-off lead of the same phase
c2	= Between (-) change-over selector contacts of different phases

Insulation distance	Voltage waveform	VVS-Y, U_m=40 kV	VVS-Y, U_m=76 kV
a	LI	200	200
	LIC	220	220
	AC	50	50
a1	LI	100	100
	LIC	110	110
	AC	20	20
b	LI	200	200
	LIC	220	220
	AC	85	85
c1	LI	400	400
	LIC	440	440
	AC	100	100
c2	LI	350	350
	LIC	385	385
	AC	140	140
f1	LI	200	350
	LIC	220	385
	AC	70	140
f2	LI	300	420

Insulation distance	Voltage waveform	VVS-Y, U_m=40 kV	VVS-Y, U_m=76 kV
f2	LIC	330	462
	AC	90	165

Table 2: Rated voltage of the internal insulation, VACUTAP® VVS-Y

LI	Full wave lightning impulse (kV, 1.2/50 µs)
LIC	Chopped wave lightning impulse (kV, 1.2/50/3...6 µs)
AC	Applied voltage (kV, 50 Hz, 1 min)

Table 3: Abbreviations for the rated insulation level

3.4.2 VACUTAP® VVS-D insulation distances

Rated voltages of the internal insulation with reversing change-over selector

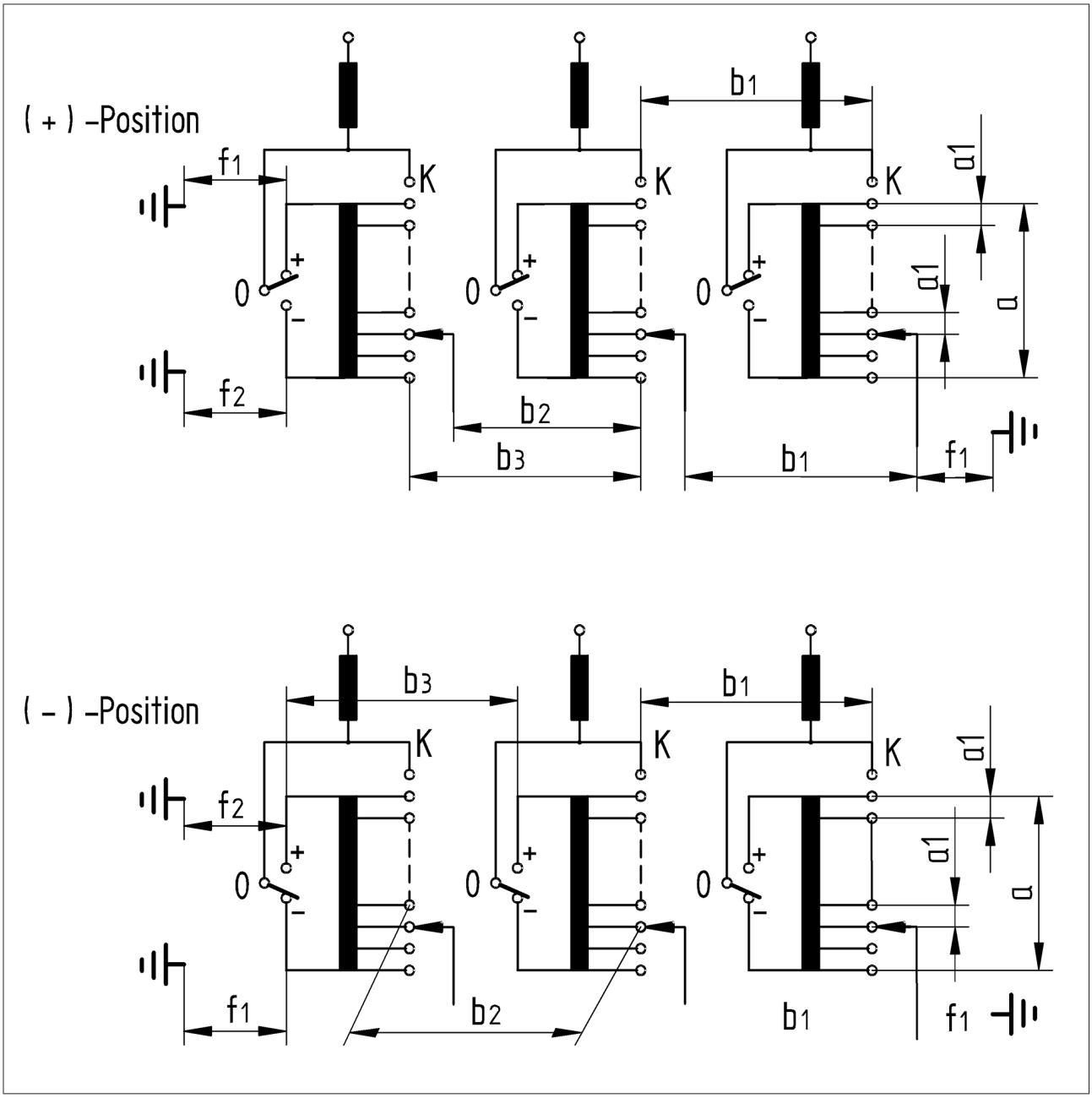


Figure 5: Rated voltages of the internal insulation with reversing change-over selector

Rated voltages of the internal insulation with coarse change-over selector

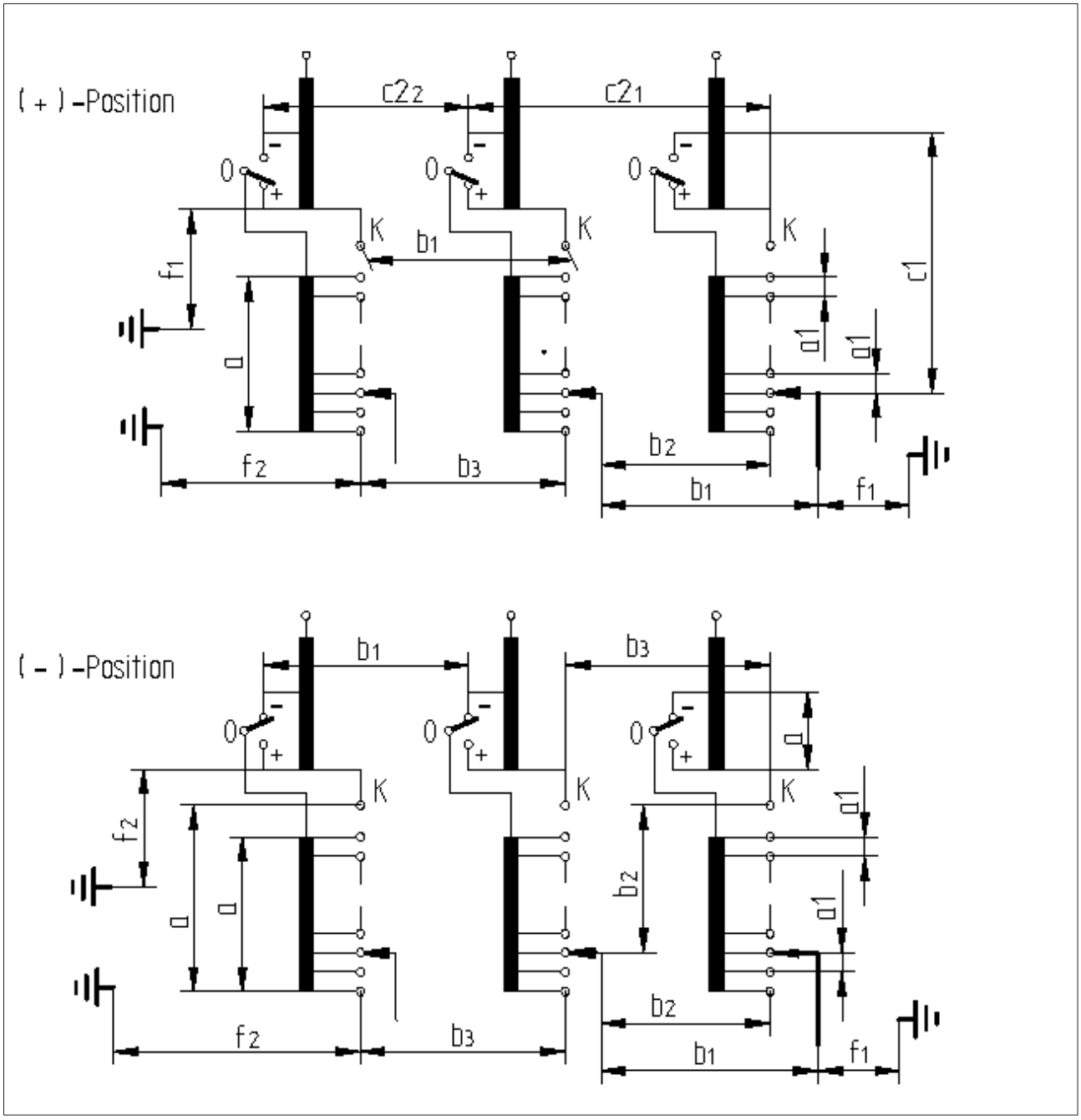


Figure 6: Rated voltages of the internal insulation with coarse change-over selector

a1	= between tap selector contacts of the winding of one tap position (connected or not connected)
a	= between beginning and end of a tap winding and, for a coarse winding model, also between beginning and end of a coarse winding.

b	= between the fine tap selector contacts of different phases and between change-over selector contacts of different phases, which are connected with the start/end of a tap winding or with a tap selector contact. The permissible voltages between the contacts for tap-changers for delta connections depend on the positions of the change-over selector and tap selector. The different permissible withstand voltages b1, b2 or b3 must therefore be noted for an insulation distance: b1 = between selected contacts for different phases b2 = from the selected contact of one phase to non-selected contacts of other phases b3 = between non-selected contacts for different phases
f	= between fine tap selector contacts and (+) change-over selector contacts to ground. Different values apply for the delta connection version: f1 = between take-off lead and ground and in (+) position of change-over selector between (+) change-over selector contacts and ground f2 = between take-off lead and ground and in (+) position of change-over selector between (+) change-over selector contacts and ground
Additionally for coarse tap selector connection in (+) position of the change-over selector:	
	c1 = from one (-) change-over selector contact to take-off lead of the same phase c2 = between (-) change-over selector contacts of different phases and from one (-) change-over selector contact to one (+) change-over selector contact of another phase. For tap-changers, different permissible values should be noted for the delta connection: c21 = between a (-) change-over selector contact and a (+) change-over selector contact of another phase. c22 = between (-) change-over selector contacts of different phases

Insulation distance	Voltage waveform	VVS-D, U_m=40 kV	VVS-D, U_m=76 kV
a	LI	200	200
	LIC	220	220
	AC	50	50
a1	LI	100	100
	LIC	110	110
	AC	20	20
b1	LI	200	350
	LIC	220	385
	AC	70	140
b2	LI	230	420
	LIC	253	462
	AC	70	165
b3	LI	300	490
	LIC	330	540

Insulation distance	Voltage waveform	VVS-D, U_m=40 kV	VVS-D, U_m=76 kV
b3	AC	90	200
c1	LI	400	400
	LIC	440	440
	AC	100	100
c21	LI	350	490
	LIC	385	540
	AC	140	165
c22	LI	350	490
	LIC	385	540
	AC	140	200
f1	LI	200	350
	LIC	220	385
	AC	70	140
f2	LI	300	420
	LIC	330	462
	AC	90	165

Table 4: Rated voltage of the internal insulation, VACUTAP® VVS-D

LI	Full wave lightning impulse (kV, 1.2/50 µs)
LIC	Chopped wave lightning impulse (kV, 1.2/50/3...6 µs)
AC	Applied voltage (kV, 50 Hz, 1 min)

Table 5: Abbreviations for the rated insulation level

3.5 Tapped winding potential connection

During its switching operation the tapped winding is briefly electrically isolated from the main winding by the reversing change-over selector or coarse change-over selector. It then adopts a potential resulting from the voltages of the adjacent windings and coupling capacities for these windings or earthed parts.

This potential shift of the tapped winding produces corresponding voltages between the deactivating change-over selector contacts because one contact is always connected to the tapped winding and the other contact is always connected to the main winding. This voltage is known as the recovery voltage U_W.

When separating the change-over selector contacts, a capacitive current has to be interrupted. This current depends on the aforementioned coupling capacities of the tapped winding. This current is known as the breaking current I_S.

The recovery voltage U_W and breaking current I_S may result in impermissible signs of discharge on the change-over selector. The permissible range of recovery voltage U_W and breaking current I_S can be seen in the following diagrams:

3.5.1 Recovery voltage and breaking current

During its switching operation the tapped winding is briefly electrically isolated from the main winding by the reversing change-over selector or coarse change-over selector. It then adopts a potential resulting from the voltages of the adjacent windings and coupling capacities for these windings or earthed parts.

This potential shift of the tapped winding produces corresponding voltages between the deactivating change-over selector contacts because one contact is always connected to the tapped winding and the other contact is always connected to the main winding. This voltage is known as the recovery voltage U_W.

When separating the change-over selector contacts, a capacitive current has to be interrupted. This current depends on the aforementioned coupling capacities of the tapped winding. This current is known as the breaking current I_S.

The recovery voltage U_W and breaking current I_S may result in impermissible signs of discharge on the change-over selector. The permissible range of recovery voltage U_W and breaking current I_S can be seen in the following diagram:

Permissible recovery voltages and take-off currents

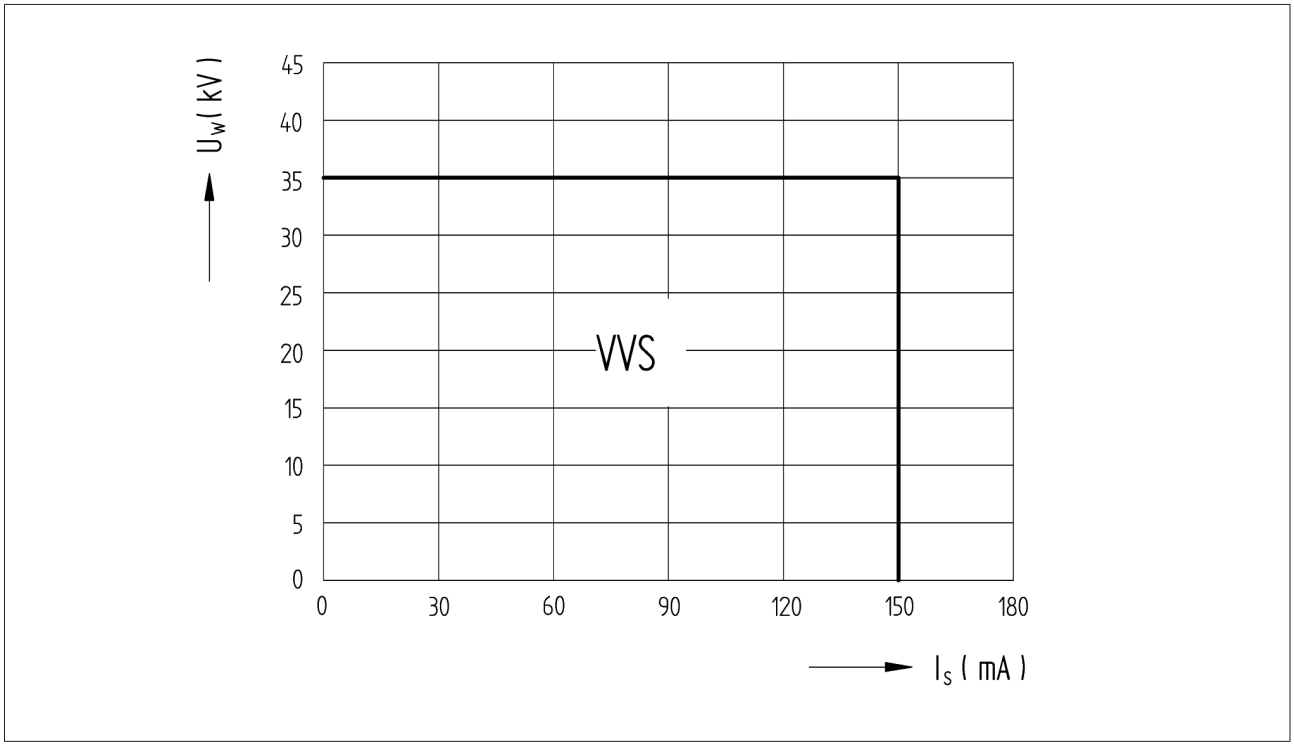


Figure 7: Recovery voltage U_w and breaking current I_s

3.6 Overload

3.6.1 Through-currents greater than the rated through-current

MR on-load tap-changers and de-energized tap-changers are suitable for all loadings of the transformer in accordance with IEC 60076-7:2005 "Loading guide for oil-immersed power transformers".

IEC 60076-7 distinguishes between three types of operation:

- Normal cyclic loading
- Long-time emergency loading
- Short-time emergency loading

The suitability of on-load tap-changers and de-energized tap-changers for the above types of power transformer operation is proven by type testing in accordance with IEC 60214-1.

MR on-load tap-changers and de-energized tap-changers are also suited to all transformer loads in accordance with IEEE Std C57.91™-2011 "IEEE Guide for Loading Mineral-Oil-Immersed Transformers and Step-Voltage-Regulators" with the following exception: Overload requirement greater than 200%.

Overload requirementsgreater than 200% may arise, e.g. for the "Short time emergency loading" type of operation with distributor transformers and must be stated in the query.

IEEE C57.91 distinguishes between four types of operation:

- Normal life expectancy loading
- Planned loading beyond nameplate rating
- Long-time emergency loading
- Short-time emergency loading

When operating with "normal cyclic loading" or "normal life expectancy loading", through-currents greater than the rated through-current may arise during a daily load cycle. If the operating conditions stated in IEC 60076-7 and IEEE C57.91 (duration and level of power during one daily cycle, transformer oil temperature etc.) are observed, this is not considered extraordinary loading but normal operation. Therefore, for the types of operation stated, particular consideration does not have to be given to the possible brief instances of through-currents greater than the rated through-current when selecting the on-load tap-changer.

3.6.2 Operation under varying operating conditions

If a transformer is operated under varying operating conditions with varying power levels (e.g. increased transformer power due to type of cooling or ambient temperature), note the following:

The rated through-current required for an on-load tap-changer must be based on the maximum transformer power as the rated power; see also IEC 60076-1:2011.

This is necessary because, due to increased power, the transformer's oil temperature is not reduced despite increased transformer cooling and therefore, contrary to the transformer, the external operating conditions of the on-load tap-changer are not improved.

It is also necessary because the transition resistors of the on-load tap-changers are designed on the basis of the maximum through-current in order to limit the switching capacity loading on the on-load tap-changer contacts to permissible values.

3.6.3 Details needed for queries relating to overload conditions

When making queries relating to overload conditions, a definition with reference to the above types of operation is needed to avoid misunderstanding. The operating conditions must be described clearly.

In the event of types of operation which cannot be defined by reference to IEC 60076-7:2005 or IEEE Std C57.91™-2011, the following details are needed:

- Through-currents and associated loading duration during one daily cycle
- Oil temperature of the transformer during one daily cycle
- Expected number of tap-change operations during loading phases of one daily cycle (for on-load tap-changer only)
- Duration of overload operation in days/weeks/months
- Frequency of overload operation, e.g. "once a year" or "rarely, only if other transformers fail".

3.7 Loading from short-circuits

The permissible loading from short-circuits is given by:

- Rated short-time current as effective value of permissible short-circuit current
- Rated peak withstand current as highest permissible peak value of short-circuit current
- Rated duration of short-circuits as permissible short-circuit duration when loaded with rated short-time current

All MR on-load tap-changers and de-energized tap-changers at least satisfy the requirements of IEC 60214-1 in terms of short-circuit strength. The permissible short-circuit duration when loaded with short-time currents lower than the rated short-time current or the permissible short-time current for durations longer than the rated duration of short-circuits can be calculated using the following equation:

$I_x^2 \cdot t_x = I_K^2 \cdot t_K$

I_K	Rated short-time current
t_K	Rated duration of short-circuits
I_x	Permissible short-time current during short-circuit duration t_x (where t_x is always greater than t_k)
t_x	Permissible short-circuit duration during loading with I_x (where I_x is always less than I_k)

Given that the dynamic loading is determined solely by the peak current, a peak current greater than the rated peak withstand current is not permitted. The rated values cannot therefore be converted to higher peak currents and short-time currents over a shorter short-circuit duration!

Short-circuit loads normally only occur very rarely when operating a transformer. This must be taken into account for applications with very frequent short-circuit loads - e.g. special test transformers - by selecting an on-load tap-changer with increased resistance to short-circuits. Details about the level and frequency of short-circuit loads expected are needed for this purpose.

4 Mechanical and design properties

This chapter contains general information about the mechanical and design properties.

4.1 Permissible pressure loading

Pressure loading may result from underpressure and overpressure. Excess pressure loading may result in leaks and malfunctions.

This chapter contains information about preventive measures and the most important protective devices. The Oil conservator for the on-load tap-changer oil chapter contains additional information about the height at which the oil conservator may be installed.

4.1.1 Pressure loading during oil filling and transport

After drying, the diverter switch oil compartment must be filled completely with oil again as soon as possible so that an impermissible amount of humidity is not absorbed from the surroundings. The on-load tap-changer oil compartment and transformer are simultaneously filled with new transformer oil under vacuum.

During oil filling, a connecting lead must be installed between connections E2 and Q for evacuating in order to simultaneously apply vacuum to on-load tap-changer oil compartment and transformer. The head and cover of the on-load tap-changer are vacuum-proof.

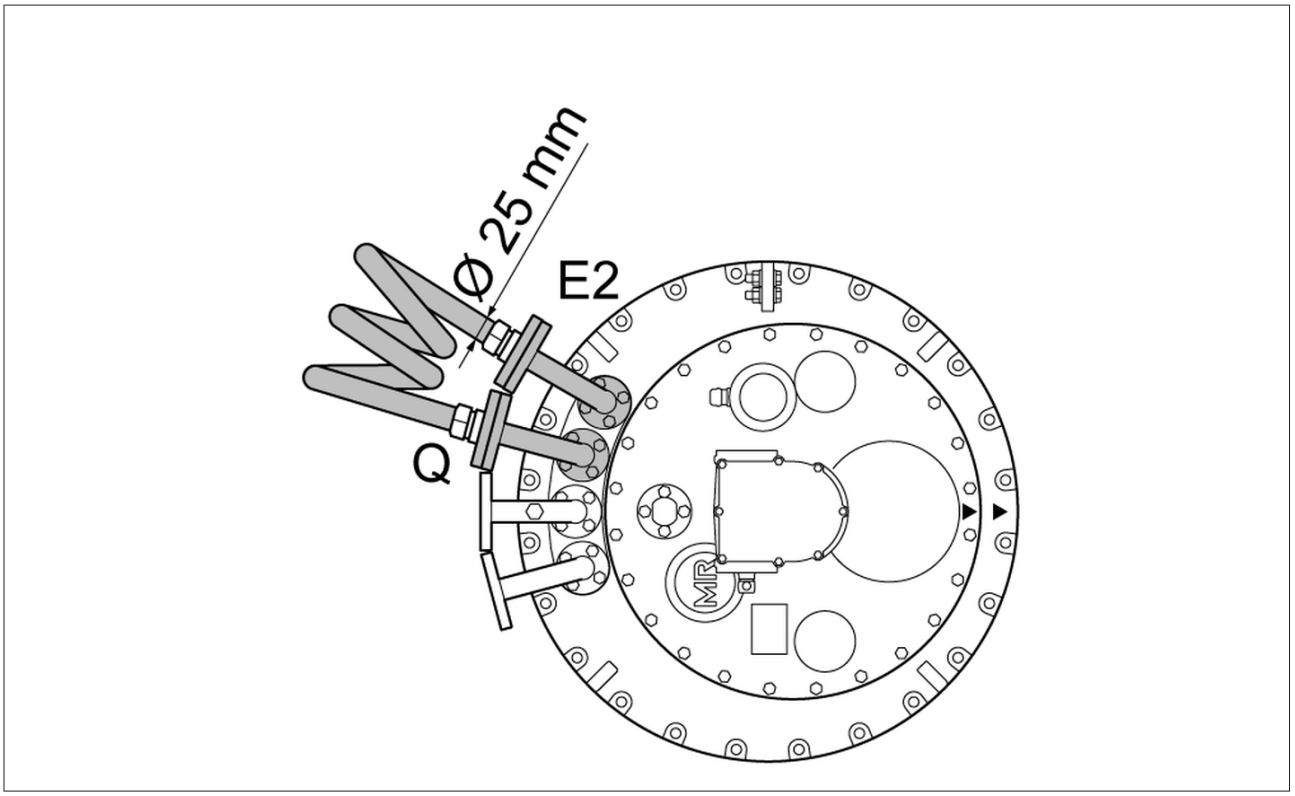


Figure 8: Connecting lead between E2 and Q.

Also if the transformer is filled with oil but stored or transported without an oil conservator, a connecting lead must be installed between the interior of the compartment and the transformer tank's oil chamber for pressure compensation. For more information on oil filling and transport, consult the relevant operating instructions.

4.1.2 Pressure loading during operation and associated protective devices

The on-load tap-changer oil compartment is pressure-resistant to 0.3 bar of continuous differential pressure (test pressure 0.6 bar). The head and on-load tap-changer head cover are vacuum-proof.

To reduce the consequences of an internal error in the on-load tap-changer, at least one protective device must be provided in accordance with IEC 60214-1.



Depending on the application area of the on-load tap-changer, varying levels of switching gas may be produced during an on-load tap-change operation.

Pressure relief device

The on-load tap-changer head covers of the MR on-load tap-changers are fitted with a rupture disk. This serves as the desired failure point for pressure relief if a pressure relief device is not used. Pressure relief devices reduce internal overpressure resulting from an internal error.

A pressure relief device is secured onto a flange on a special on-load tap-changer head cover. It comprises a housing and a sealing flap subject to spring tension with signaling contacts.

The pressure relief device must be looped into the circuit breaker's tripping circuit. When the protective device is tripped, the transformer must be immediately de-energized by the circuit breaker.

When the device's permissible tripping pressure is exceeded, the cover lifts up and the seal opens. When pressure falls below the tripping level, the device closes again. The installation height of the oil conservator must be taken into account when designing pressure relief devices.

Oil flow-controlled relay

The on-load tap-changer is fitted with an oil flow-controlled relay as standard.

The protective relay is tripped when the specified speed of oil flow from the on-load tap-changer head to the oil conservator is exceeded due to a fault. The flowing oil actuates the flap valve which tips over into the OFF position. A contact is thereby actuated which trips the circuit breaker and de-energizes the transformer.

The protective relay and additional protective devices must be looped into the circuit breaker's tripping circuit. When the protective device is tripped, the transformer must be immediately de-energized by the circuit breaker.

4.2 Oil conservator for on-load tap-changer oil

This chapter describes the special conditions for on-load tap-changers which need to be taken into account for the oil conservator's installation height and sizing.

The hydrostatic pressure of the insulating oil may restrict function and seal integrity if the installation height limits are not observed. More information about pressure can be found in the Permissible pressure loading chapter.

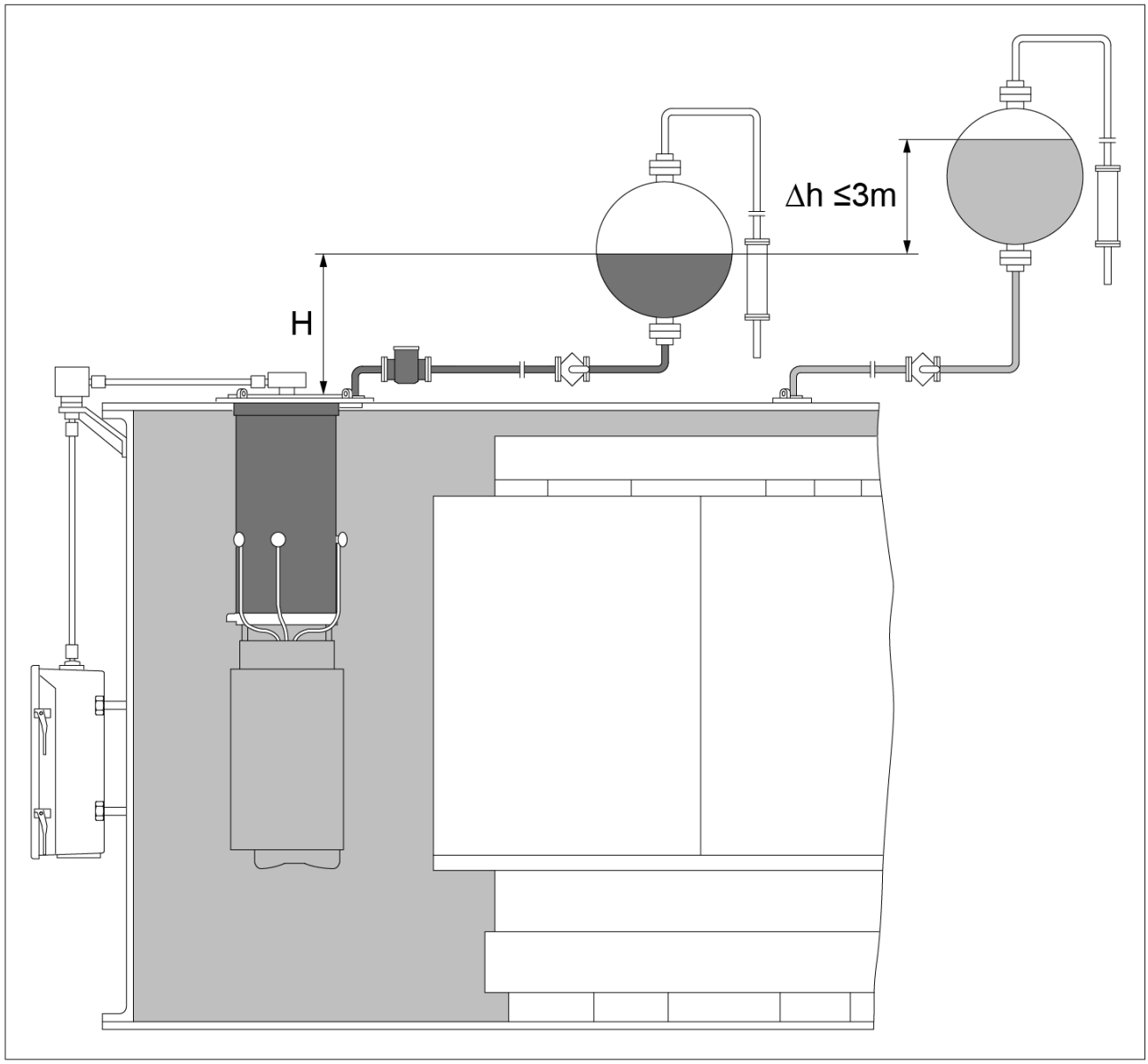


Figure 9: Overview of oil supply

Δh	Difference in height between oil levels in oil conservators
H	Height of oil level in on-load tap-changer's oil conservator above the on-load tap-changer head cover

4.2.1 Height of the oil conservator

The permissible heights for the oil conservators of the on-load tap-changer and transformer must be observed. These ensure:

- seal integrity of on-load tap-changer oil compartment in relation to surroundings and transformer
- correct function (e.g. switching process) of on-load tap-changer and other pressure-dependent devices

The standard on-load tap-changer model is designed for a maximum oil conservator height H_{max} of **up to 5 m**. To calculate this height, the distance between the maximum oil level in the oil conservator and the top edge of the on-load tap-changer head cover must be determined.

If the height H_{max} of the oil level in the on-load tap-changer's oil conservator is more than 5 m above the on-load tap-changer head cover, this must be stated in the order so that an appropriate product variant can be selected.

For VACUTAP® on-load tap-changers at installation heights H_{NHN} of more than 2,000 m above sea level, the maximum permissible height H_{max} of the oil conservator is increased by the minimum distance H_{max} between the oil level and on-load tap-changer head cover in accordance with section Installation height above sea level.

Difference in height Δh between oil levels in on-load tap-changer and transformer

If the oil conservators for the on-load tap-changer and transformer are in separate locations, the difference in height Δh between the oil levels may be a **maximum of 3 m**.

If the on-load tap-changer and transformer share an oil conservator (with or without divider), this difference is not normally reached. With a shared oil conservator, the difference in height can then be ignored.

4.2.2 Installation height above sea level

VACUTAP® oil-insulated on-load tap-changer

VACUTAP® oil-insulated on-load tap-changers with an open oil conservator are released without limitations up to an installation height H_{NHN} of 2,000 m above sea level. Above 2,000 m, a minimum height must be observed for the oil conservator.

The installation height of the oil conservator is determined by the distance H_{min} between the upper edge of the on-load tap-changer head cover and the oil level in the oil conservator.

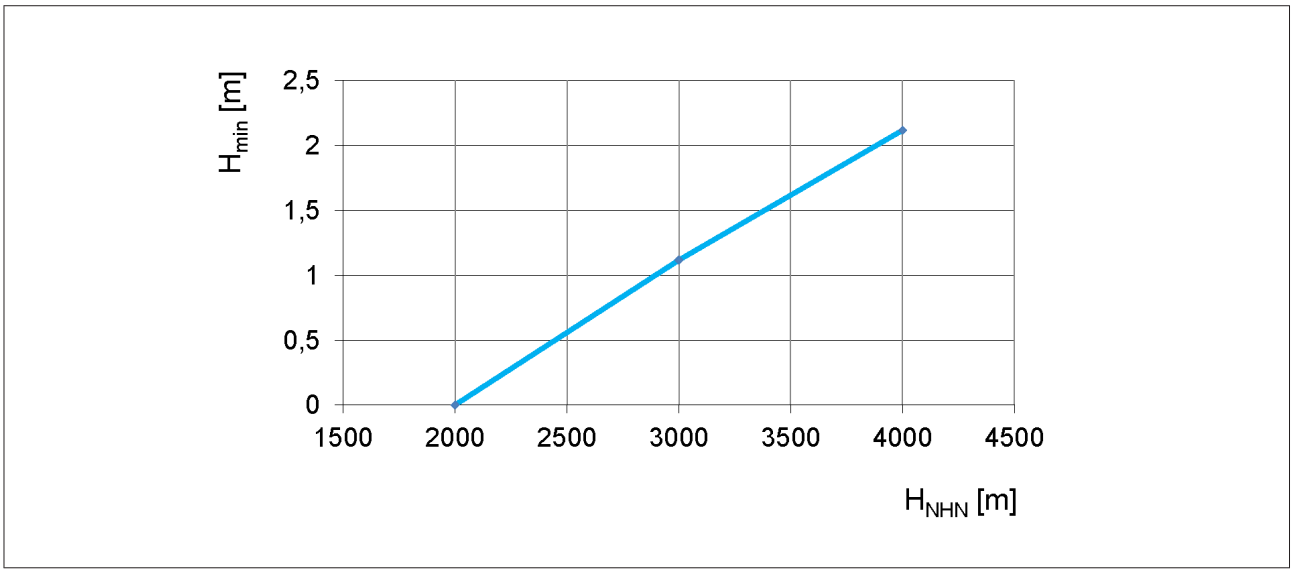


Figure 10: Minimum distance H_{min} between oil level and on-load tap-changer head cover

H_{min}	Distance between oil level in the oil conservator and upper edge of the on-load tap-changer head cover.
H_{NHN}	Installation height above sea level

For VACUTAP® on-load tap-changers at installation heights H_{NHN} of more than 2,000 m above sea level, the maximum permissible height of the oil conservator (in accordance with section Height of oil conservator) is increased by this minimum distance H_{min} between the oil level and on-load tap-changer head cover.

Example:

For an installation height H_{NHN} of 2,500 m above sea level, the maximum permissible height $H_{max}$of the oil conservator is as follows:

$H_{max(2500m)} = H_{max(0m)} + H_{min} = 5\ m + 0.5\ m = 5.5\ m.$

For installation heights H_{NHN} of more than 4,000 m or other applications, e.g. hermetic sealing, please contact Maschinenfabrik Reinhausen GmbH (MR).

4.2.3 Minimum oil conservator volume

The maximum expansion of the on-load tap-changer oil must be taken into account when sizing. This provides the useable volume required that must be available inside the oil conservator.

The entire oil volume in the on-load tap-changer's oil supply must be taken into account for the oil fill. The stated minimum fill volume within the on-load tap-changer oil compartment is part of this and relates to oil expansion at 20 °C.

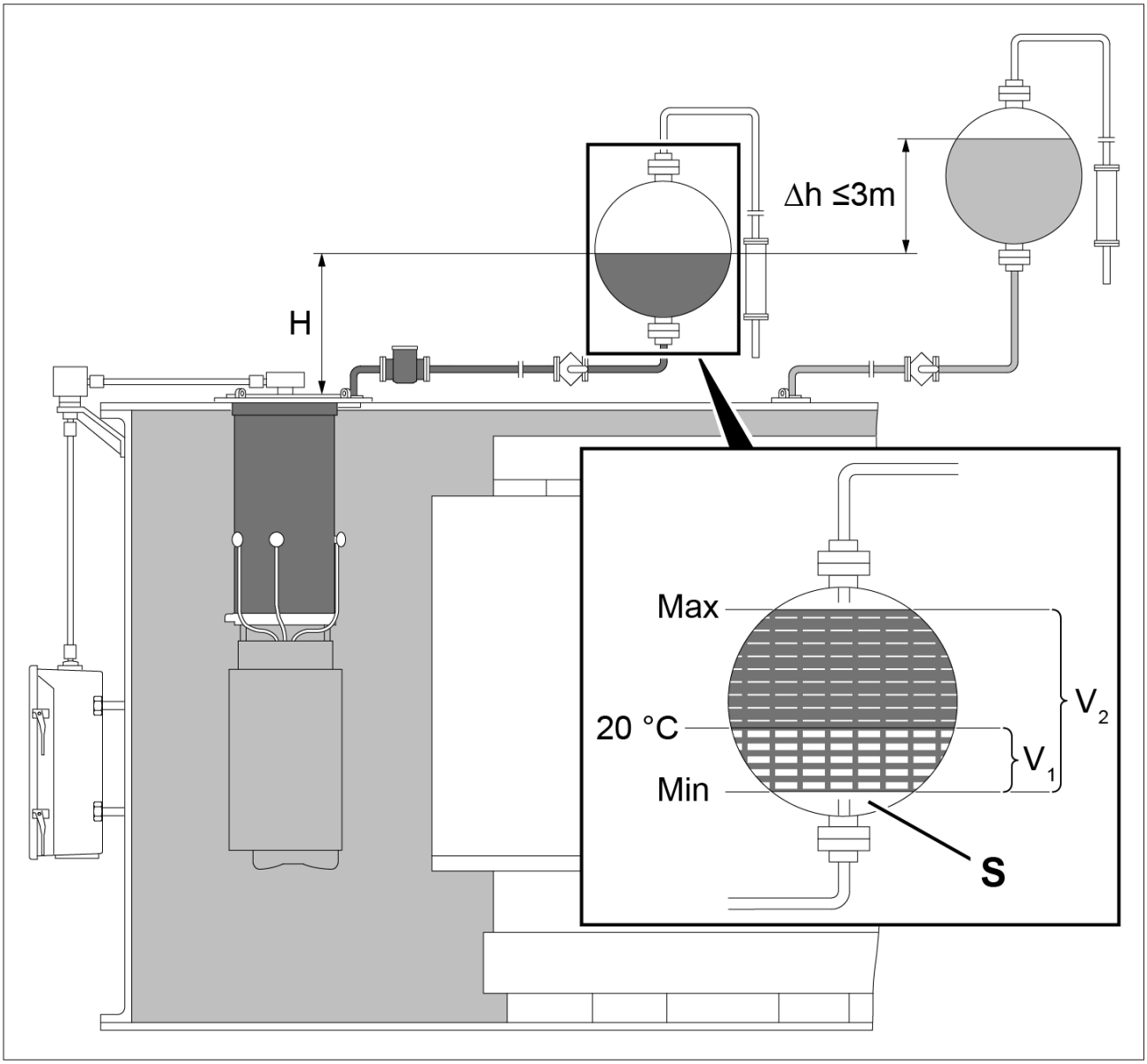
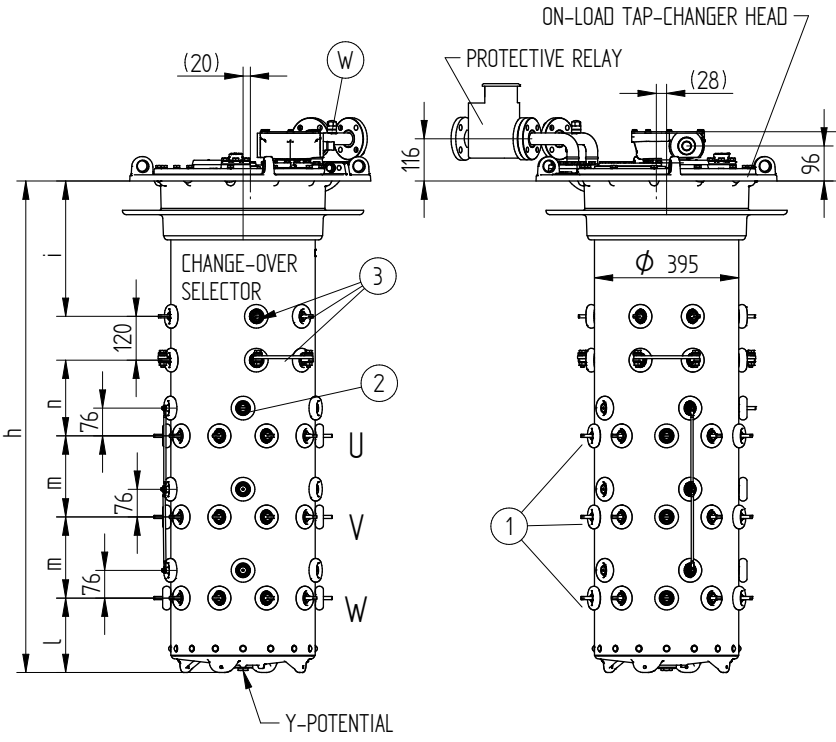


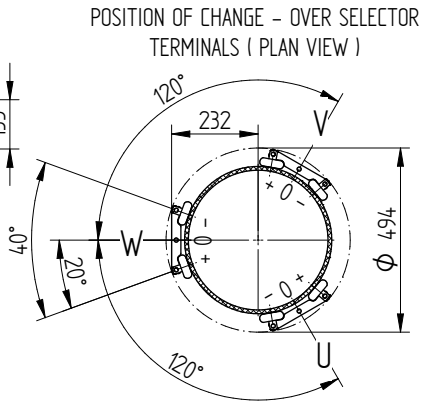
Figure 11: Expansion volume and minimum fill volume

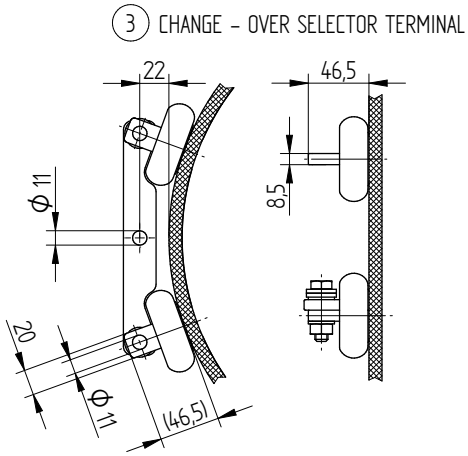
S	Sump in oil conservator
V_1	Minimum fill volume in oil conservator at 20 °C
V_2	Expansion volume of on-load tap-changer oil = minimum useable volume of oil conservator

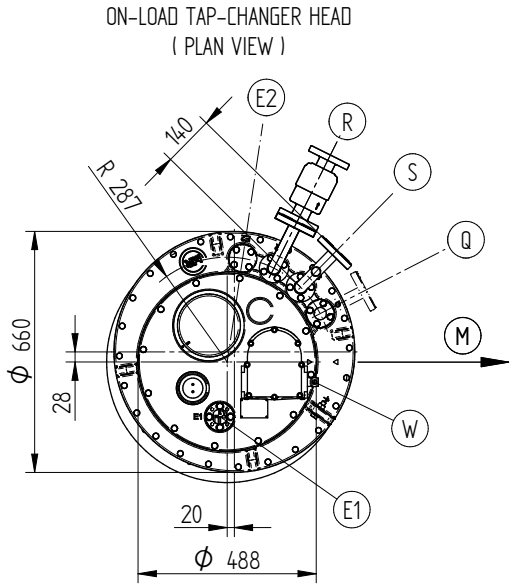
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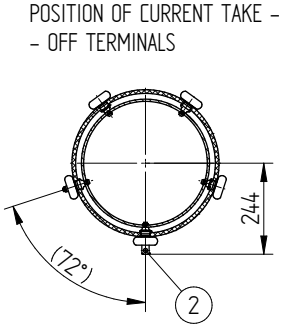
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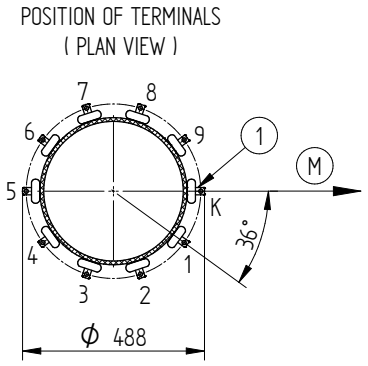


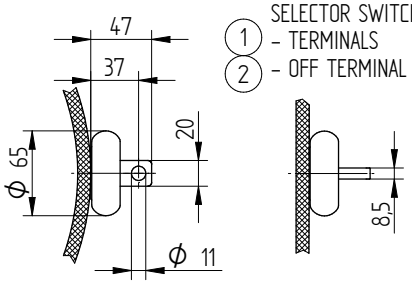












(E1) BLEEDING OF ON-LOAD TAP-CHANGER HEAD

(E2) BLEEDING OF TRANSFORMER TANK

(Q) CONNECTING FLANGE FOR OIL RETURN (WITH FILTER UNIT ONLY)

(R) CONNECTING FLANGE FOR PROTECTIVE RELAY

(S) CONNECTING FLANGE FOR SUCTION PIPE

(M) MARK (DRIVE SIDE)

(W) DRIVE SHAFT

FOR BINDING DESIGNATIONS OF TERMINALS AND PHASES REFER TO THE CONNECTION DIAGRAM OF THE ON-LOAD TAP-CHANGER

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	m	222	
	l	204	

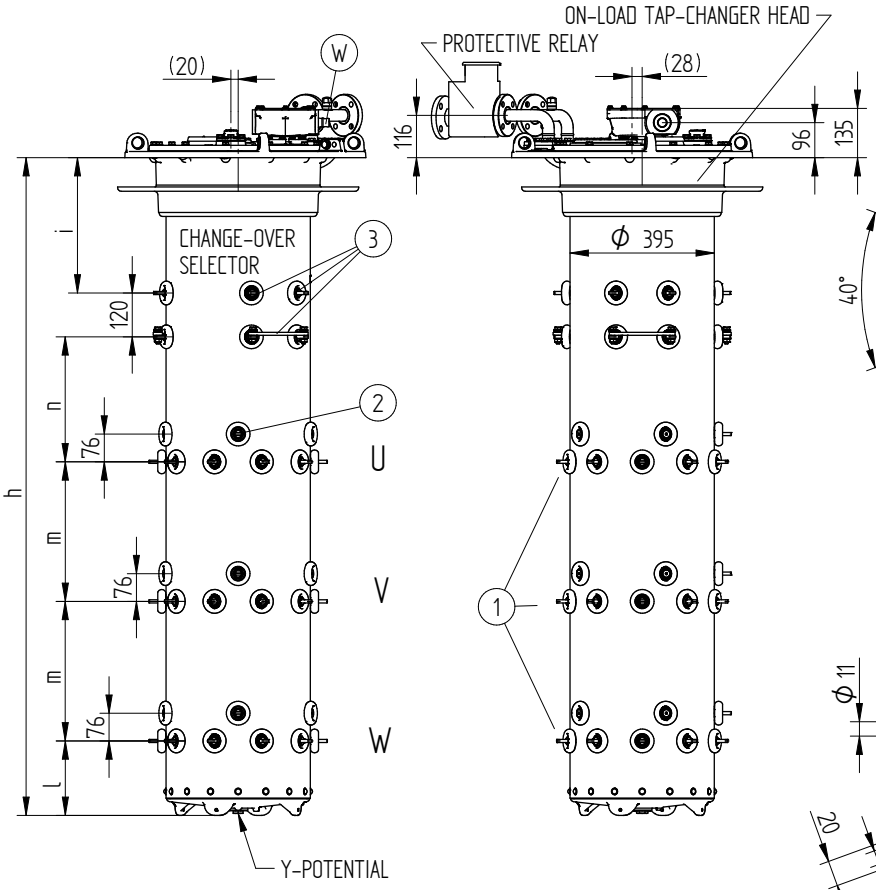
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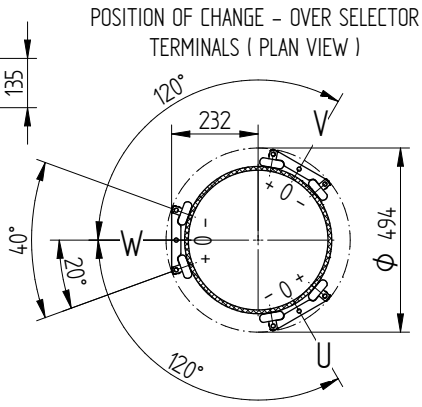
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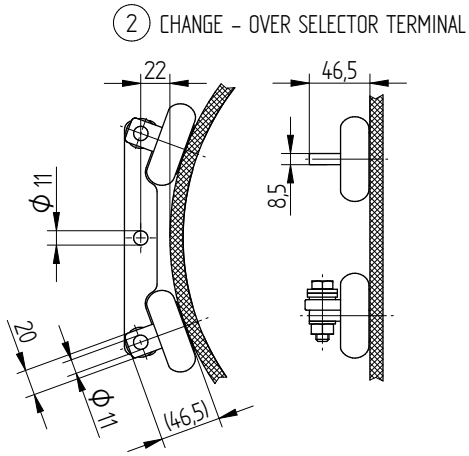
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VVS III Y - W/G / MODEL WITH UPPER GEAR UNIT
DIMENSION DRAWING

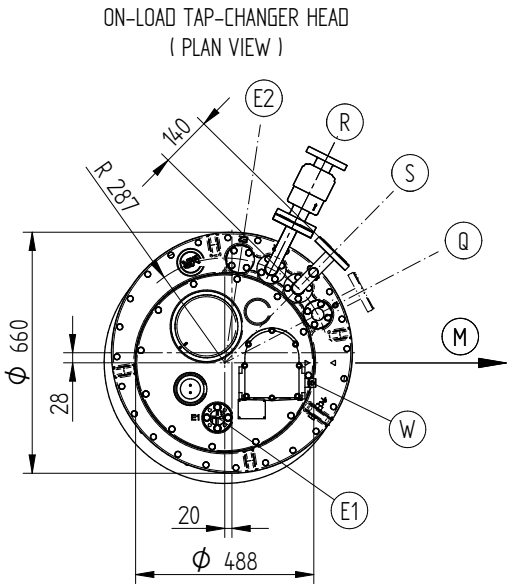
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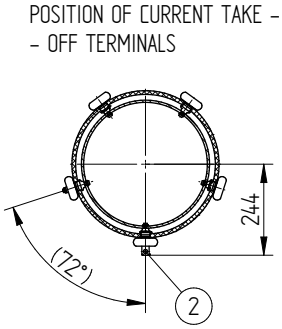
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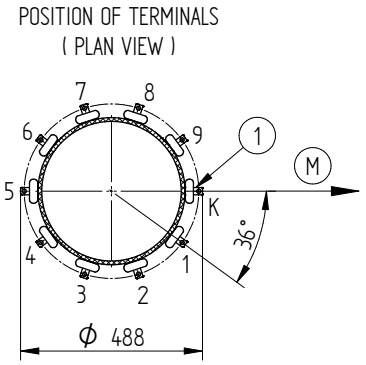


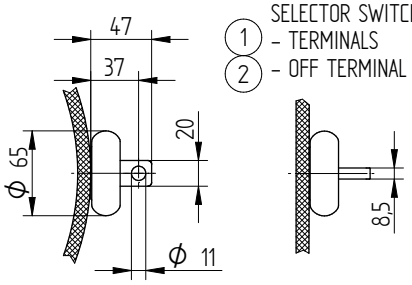












(E1) BLEEDING OF ON-LOAD TAP-CHANGER HEAD

(E2) BLEEDING OF TRANSFORMER TANK

(Q) CONNECTING FLANGE FOR OIL RETURN (WITH FILTER UNIT ONLY)

(R) CONNECTING FLANGE FOR PROTECTIVE RELAY

(S) CONNECTING FLANGE FOR SUCTION PIPE

(M) MARK (DRIVE SIDE)

(W) DRIVE SHAFT

FOR BINDING DESIGNATIONS OF TERMINALS AND PHASES REFER TO THE CONNECTION DIAGRAM OF THE ON-LOAD TAP-CHANGER

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	m	382	
	l	204	

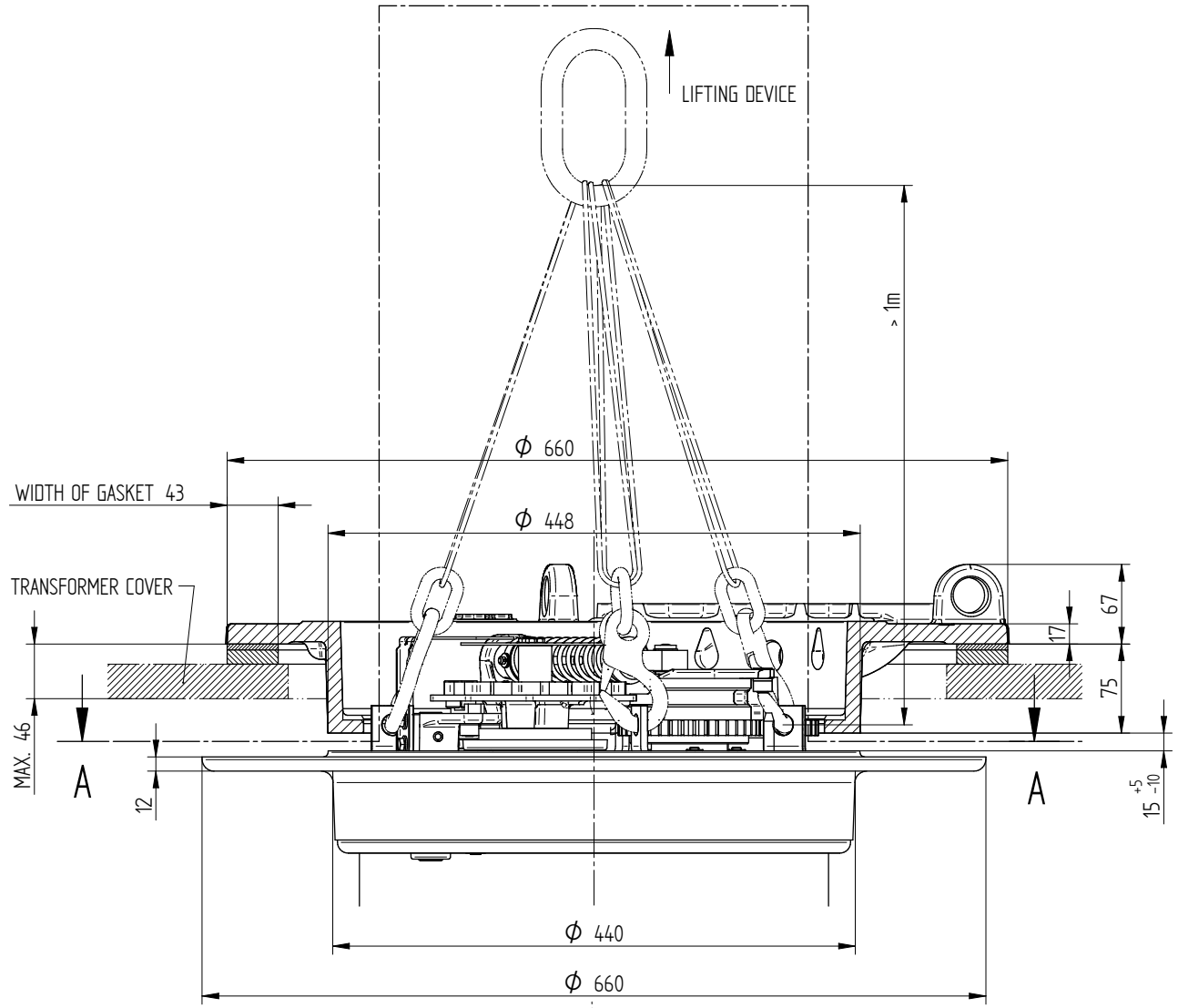
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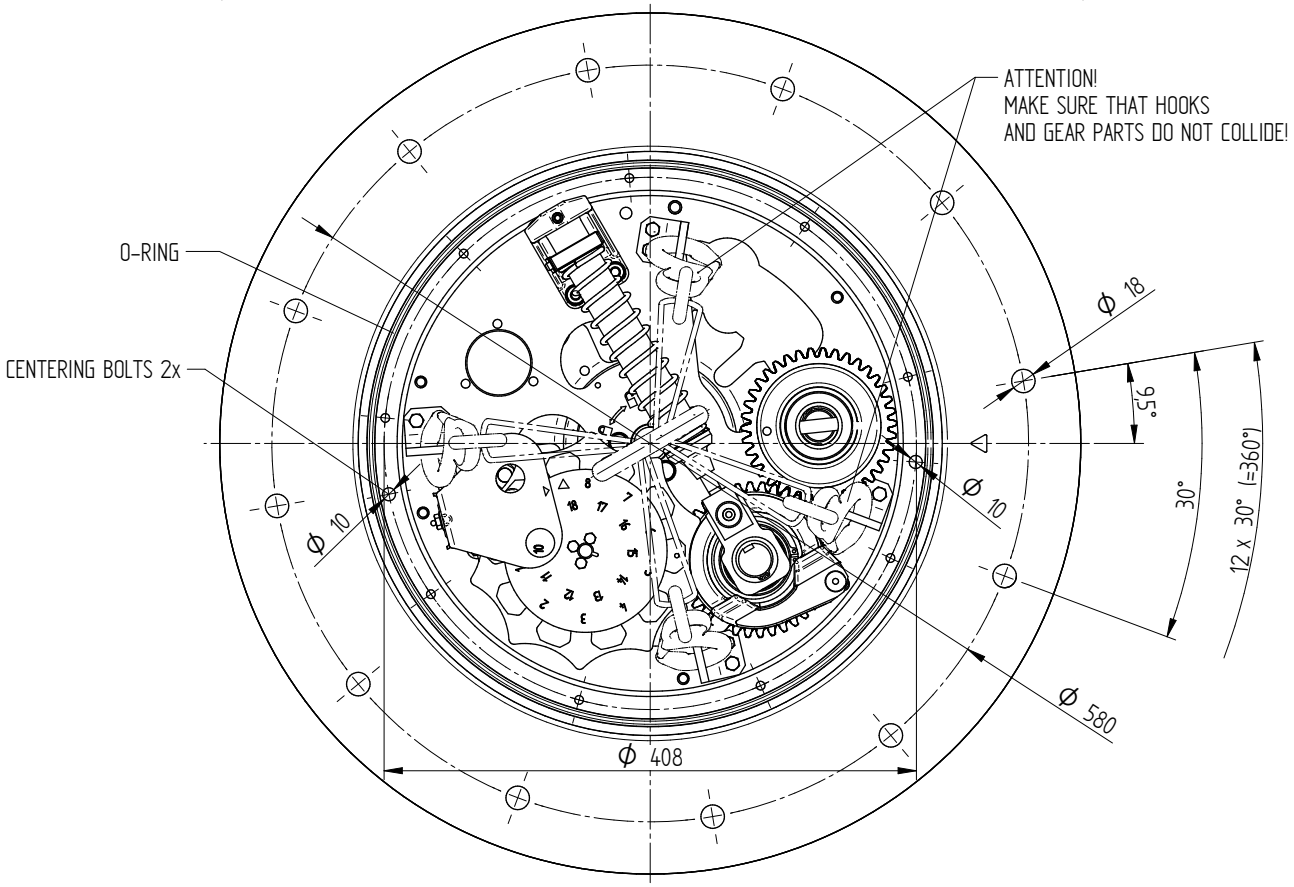
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ON-LOAD TAP-CHANGER VACUTAP® VVS®
VVS III 400 D - W/G / MODEL WITH UPPER GEAR UNIT
DIMENSION DRAWING

SERIAL NUMBER
-
MATERIAL NUMBER 101177440E
SHEET 1/1
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DIMENSION DRAWING

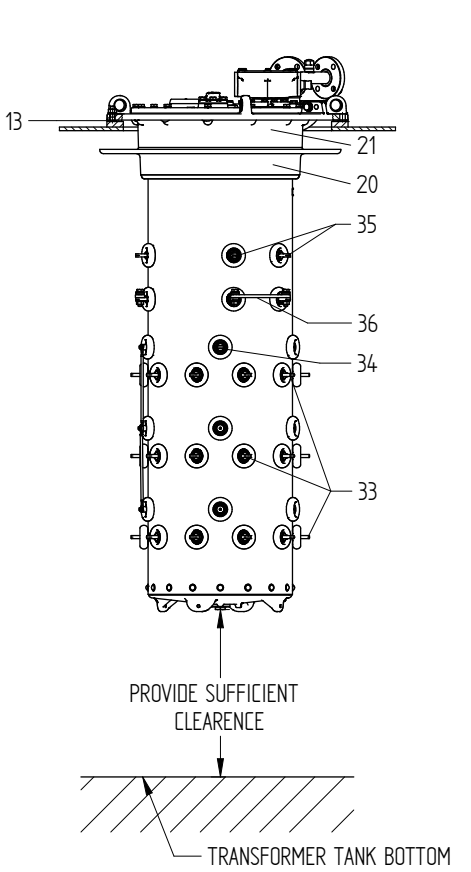
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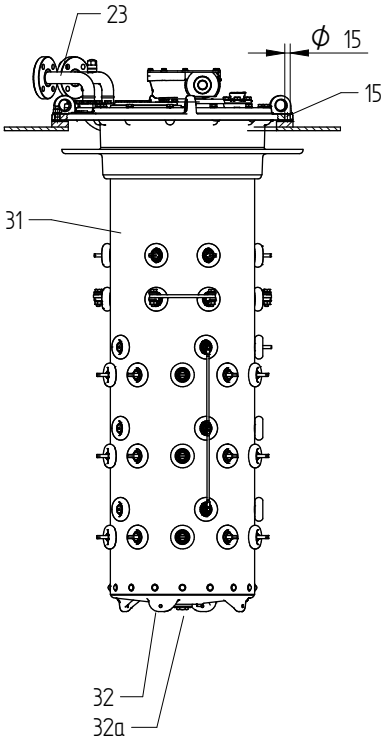
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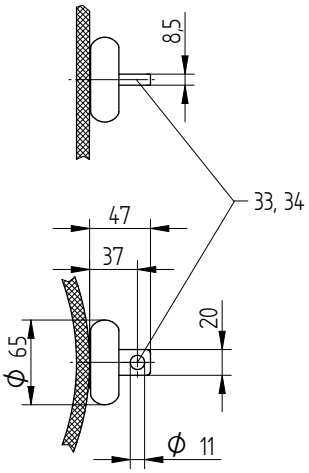
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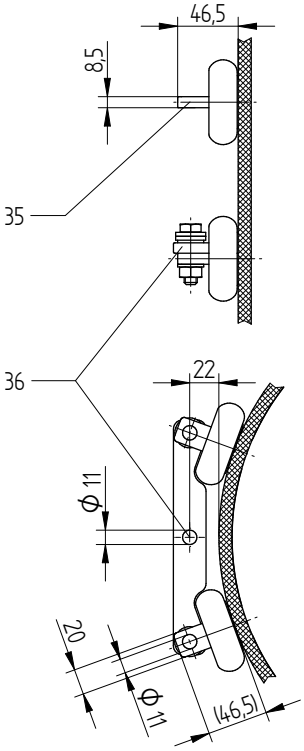
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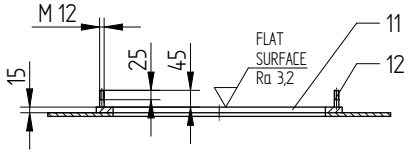
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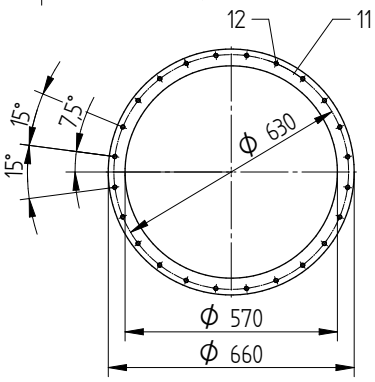


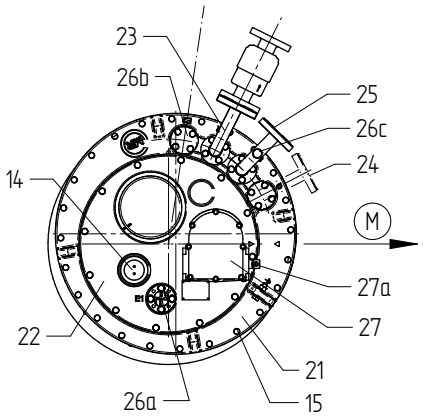












(M) = MARK (DRIVE SIDE)

11 – MOUNTING FLANGE OF TRANSFORMER COVER
12 – FIXING BOLT M 12
13 – GASKET FOR ON-LOAD TAP-CHANGER HEAD
14 – INSPECTION WINDOW FOR POS. HEAD
15 – THROUGH – HOLE ⌀15

20 – ON-LOAD TAP-CHANGER HEAD – BOTTOM PART
21 – ON-LOAD TAP-CHANGER HEAD – TOP PART
22 – COVER OF ON-LOAD TAP-CHANGER HEAD
23 – PIPE CONNECTION "R" FOR PROTECTIVE RELAY
24 – PIPE CONNECTION "Q" FOR RETURN PIPE (WITH OIL FILTER UNIT ONLY)
25 – PIPE CONNECTION "S" FOR SUCTION PIPE
26a – BLEEDER SCREW OF ON-LOAD TAP-CHANGER HEAD COVER
26b – BLEEDER SCREW FOR TRANSFORMER OIL
26c – BLEEDER SCREW FOR SUCTION PIPE
27 – UPPER GEAR UNIT WITH DRIVE SHAFT 27a

31 – SELECTOR SWITCH OIL COMPARTMENT
32 – BOTTOM OF OIL COMPARTMENT WITH KEROSENE SCREW 32a
33 – SELECTOR SWITCH TERMINAL
34 – OUTPUT TERMINAL OR NEUTRAL TERMINAL
35 – CHANGE – OVER SELECTOR TERMINAL " + " AND " – "
36 – CHANGE – OVER SELECTOR TERMINAL " 0 "

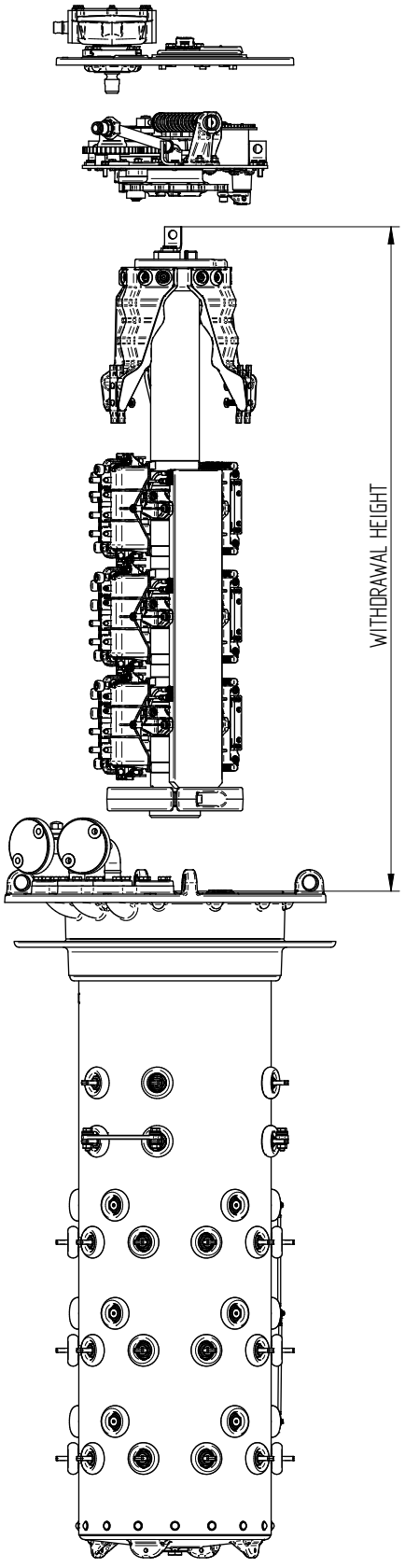
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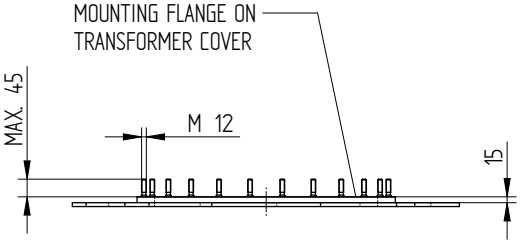
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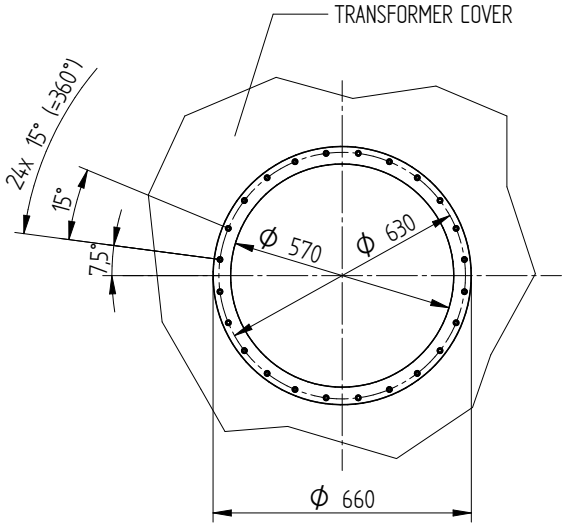
ON-LOAD TAP-CHANGER VACUTAP® VVS®
MODEL WITH UPPER GEAR UNIT-
INSTALLATION DRAWING

SERIAL NUMBER
–
MATERIAL NUMBER 101168880E
SHEET 1/1
A1

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ON-LOAD TAP-CHANGER	WITHDRAWAL HEIGHT
V V S III Y – 76 kV	1364 MM
V V S III Y – 40 kV	
V V S III D – 76 kV	1819 MM
V V S III D – 40 kV	

⚠ THE WITHDRAWAL HEIGHT CORRESPONDS TO THE CLEARANCE BETWEEN THE COVER FLANGE AND THE LIFTING DEVICE OF THE OLTC INSERT, WHILE THE MOUNTING ROD IS INSERTED. ADDITIONAL FIXING PADS, WHICH MAY BE NECESSARY FOR AFFIXING THE OLTC INSERT, HAVE TO BE CONSIDERED TOO.

	DATE	NAME	DOCUMENT NO.	
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CHKD.	20.11.2018	STREMPEL	CHANGE NO.	SCALE
STAND.	20.11.2018	KLEYN	1091025	1:1

DIMENSION IN mm EXCEPT AS NOTED

MR

ON-LOAD TAP-CHANGER VACUTAP® VVS®

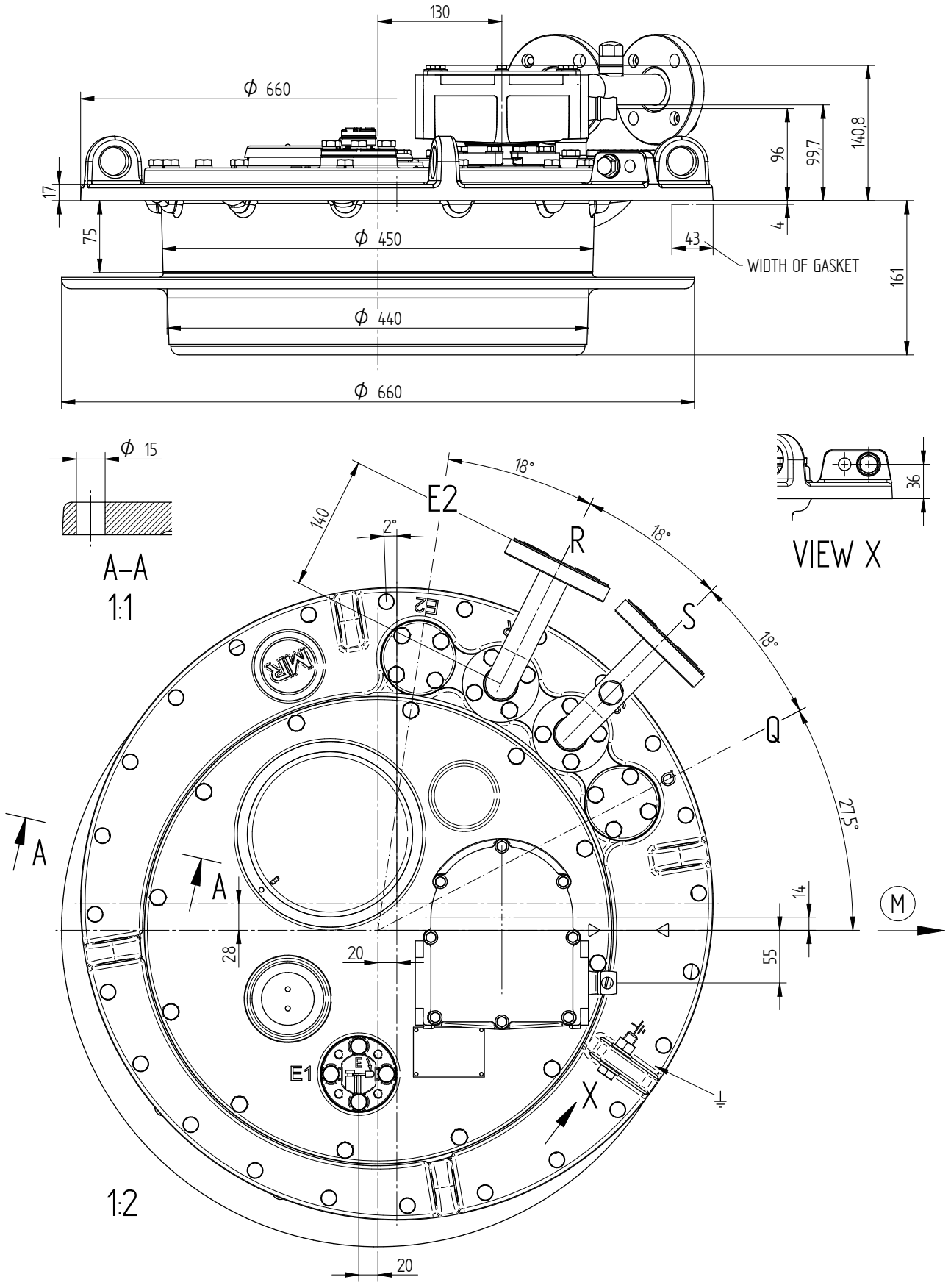
INSTALLATION DRAWING

SERIAL NUMBER
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MATERIAL NUMBER
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SHEET
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E1 = BLEEDING FACILITY FOR TAP-CHANGER HEAD
E2 = BLEEDING FACILITY FOR SPACE UNDER THE HEAD OUTSIDE THE TAP-CHANGER OIL COMPARTMENT
R = CONNECTION FOR PROTECTIVE RELAY (EXCHANGEABLE WITH CONNECTION Q)
Q = CONNECTION FOR OIL RETURN (ONLY FOR OIL FILTER) — CONNECTIONS ORIENTABLE THROUGH 360°
S = CONNECTION FOR SUCTION PIPE
⏚ = EARTH CONNECTION M 12
Ⓜ = MARK (DRIVE SIDE)

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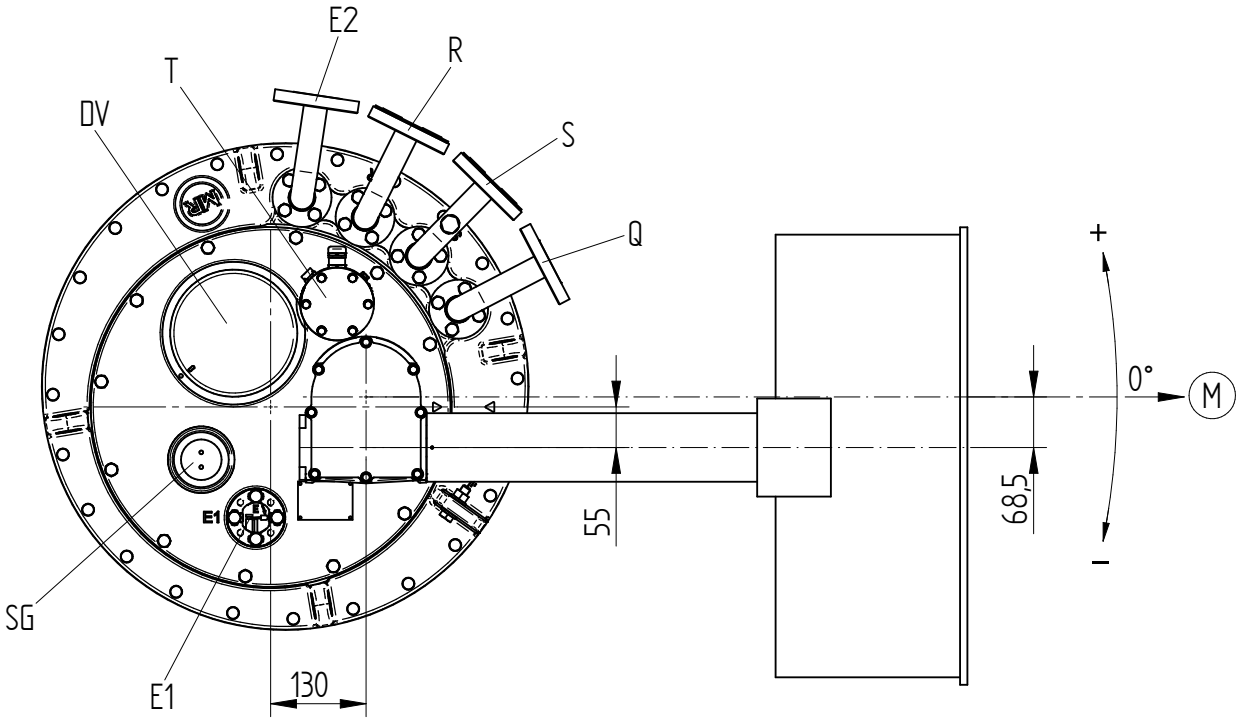
DIMENSION IN mm EXCEPT AS NOTED

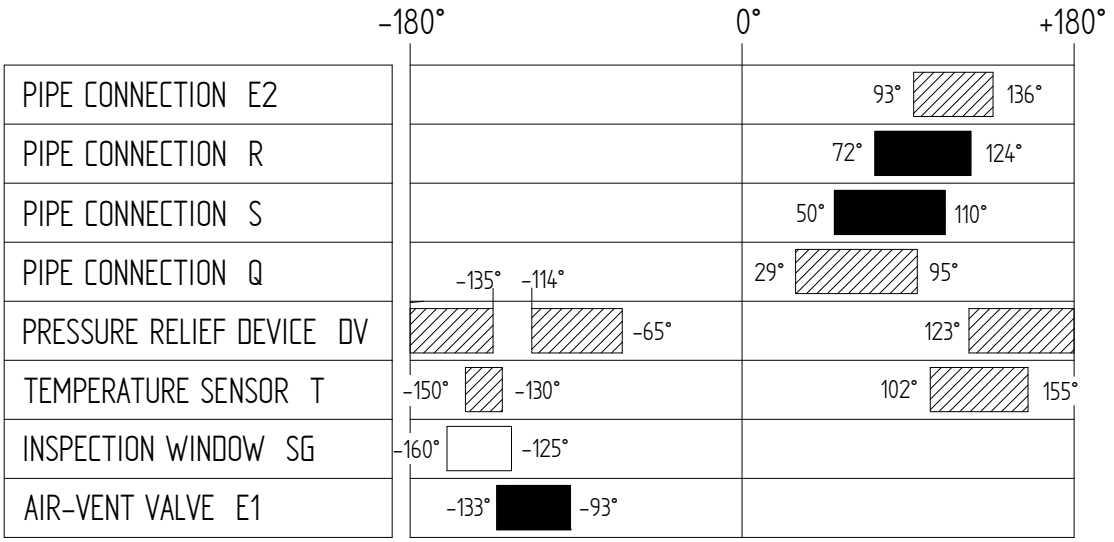
ON-LOAD TAP-CHANGER VACUTAP® VVS®
ON-LOAD TAP-CHANGER HEAD
MODEL WITH UPPER GEAR UNIT

SERIAL NUMBER
–
MATERIAL NUMBER 101168850E
SHEET 1/1

A1

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LIMITATION OF THE SWIVEL RANGE THROUGH PIPE CONNECTIONS R AND S AND AIR-VENT VALVE E1

LIMITATION OF THE SWIVEL RANGE THROUGH OPTIONAL EXISTING PIPE CONNECTIONS Q , E2 AND PRESSURE RELIEF DEVICE DV AND TEMPERATURE SENSOR T

SWIVEL RANGE POSSIBLE, BUT THE INSPECTION WINDOW SG IS NOT VISIBLE

(M) – MARK (DRIVE SIDE)

SWIVELLING RANGE

THE ON-LOAD TAP-CHANGER CAN BE TURNED AROUND ITS AXLE AND THE UPPER GEAR UNIT SWIVELLED CONTINUOUSLY TO ALLOW FOR EASY ADAPTION TO THE TRANSFORMER.
THE POSSIBLE SWIVELLING RANGES OF THE UPPER GEAR UNIT ARE SHOWN ABOVE.

	DATE	NAME	DOCUMENT NO.	
DFTR.	06.06.2018	STREMPEL	SED 5985385 001 00	
CHKD.	06.06.2018	STREMPEL	CHANGE NO.	SCALE
STAND.	07.06.2018	PRODASTSCHUK	1087254	1:5

DIMENSION IN mm EXCEPT AS NOTED

MR

ON-LOAD TAP-CHANGER VACUTAP® VVS®
SWIVELLING RANGE FOR UPPER GEAR UNIT, SHAFT ON RIGHT SIDE – HEAD DESIGN FOR SDS–

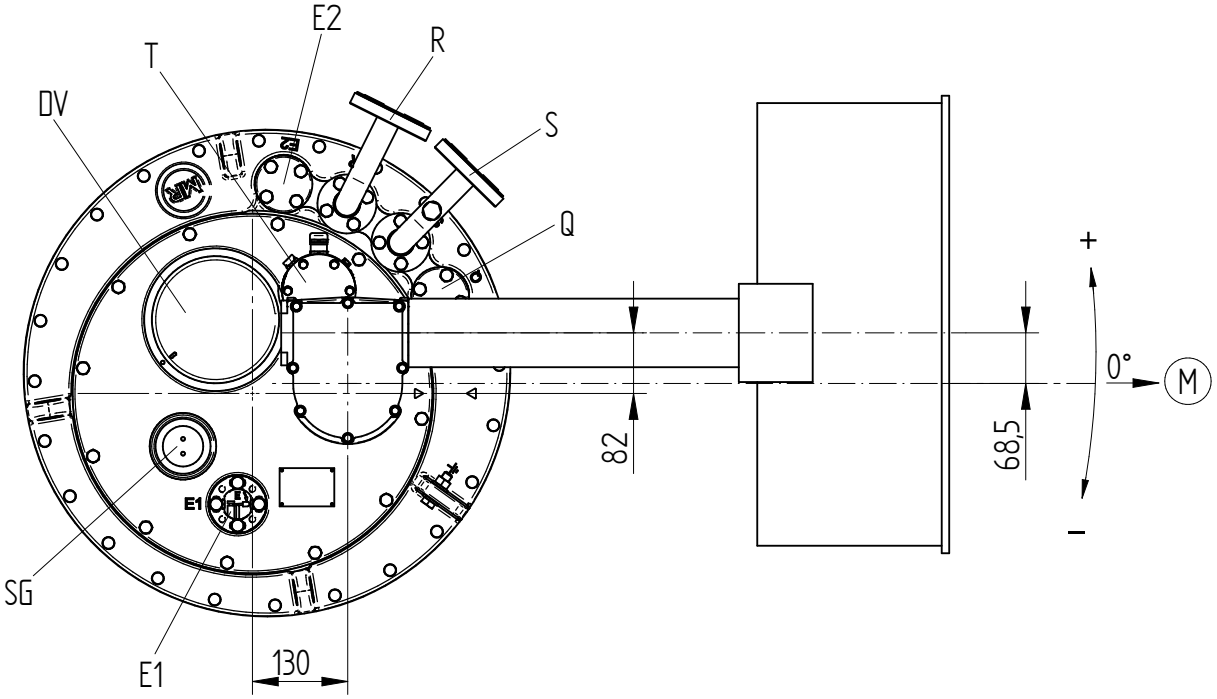
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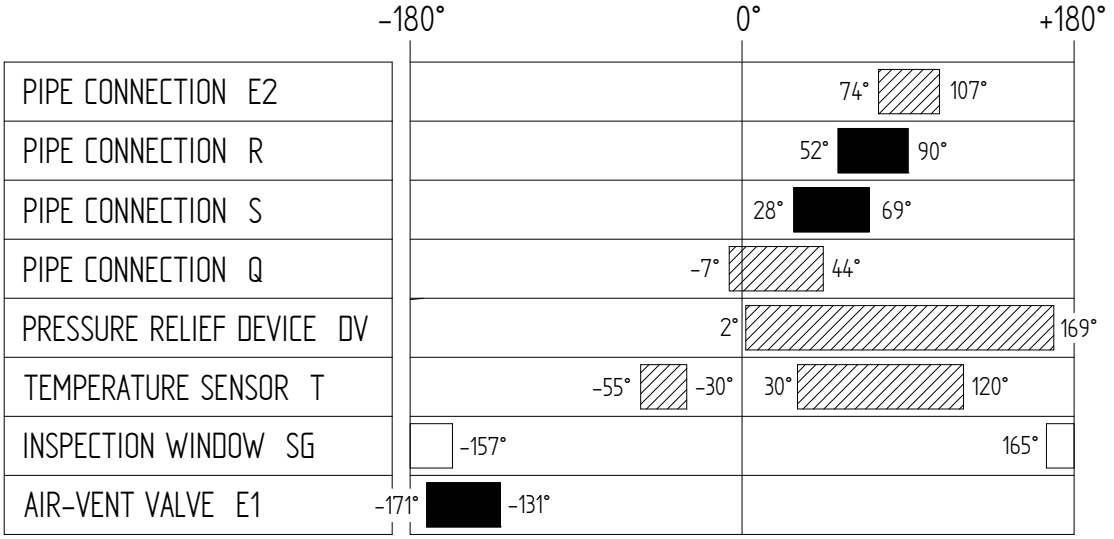
MATERIAL NUMBER 101168460E

SHEET 1 / 1

A2

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LIMITATION OF THE SWIVEL RANGE THROUGH PIPE CONNECTIONS R AND S AND AIR-VENT VALVE E1

LIMITATION OF THE SWIVEL RANGE THROUGH OPTIONAL EXISTING PIPE CONNECTIONS Q , E2 AND PRESSURE RELIEF DEVICE DV AND TEMPERATURE SENSOR T

SWIVEL RANGE POSSIBLE, BUT THE INSPECTION WINDOW SG IS NOT VISIBLE

(M) - MARK (DRIVE SIDE)

SWIVELLING RANGE

THE ON-LOAD TAP-CHANGER CAN BE TURNED AROUND ITS AXLE AND THE UPPER GEAR UNIT SWIVELLED CONTINUOUSLY TO ALLOW FOR EASY ADAPTION TO THE TRANSFORMER.
THE POSSIBLE SWIVELLING RANGES OF THE UPPER GEAR UNIT ARE SHOWN ABOVE.

	DATE	NAME	DOCUMENT NO.	
DFTR.	06.06.2018	STREMPEL	SED 586336 001 00	
CHKD.	06.06.2018	STREMPEL	CHANGE NO.	SCALE
STAND.	07.06.2018	PRODASTSCHUK	1087254	1:5

DIMENSION IN mm EXCEPT AS NOTED

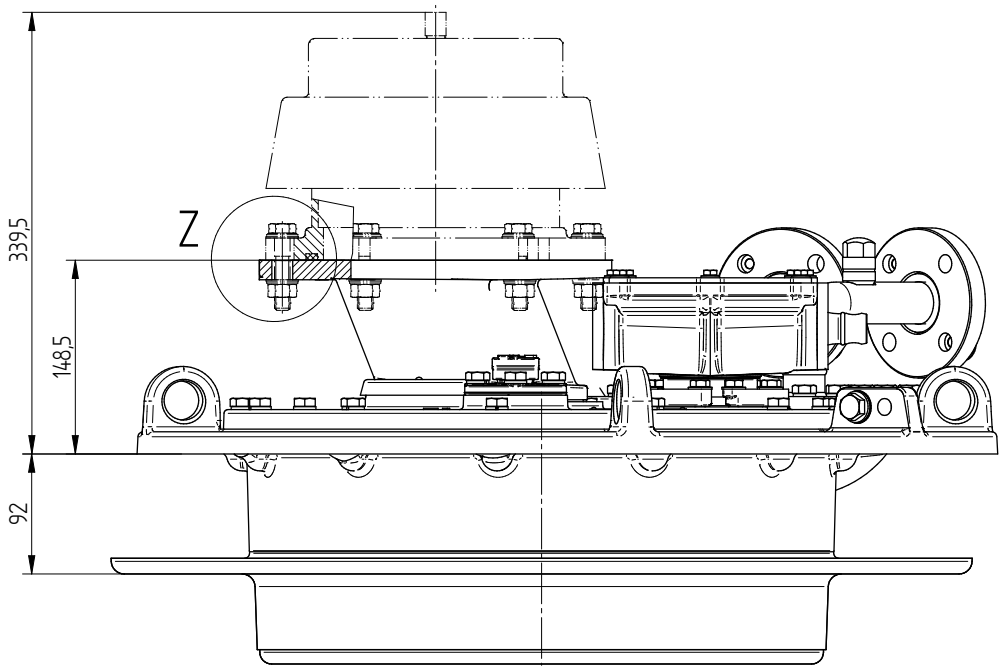
MR

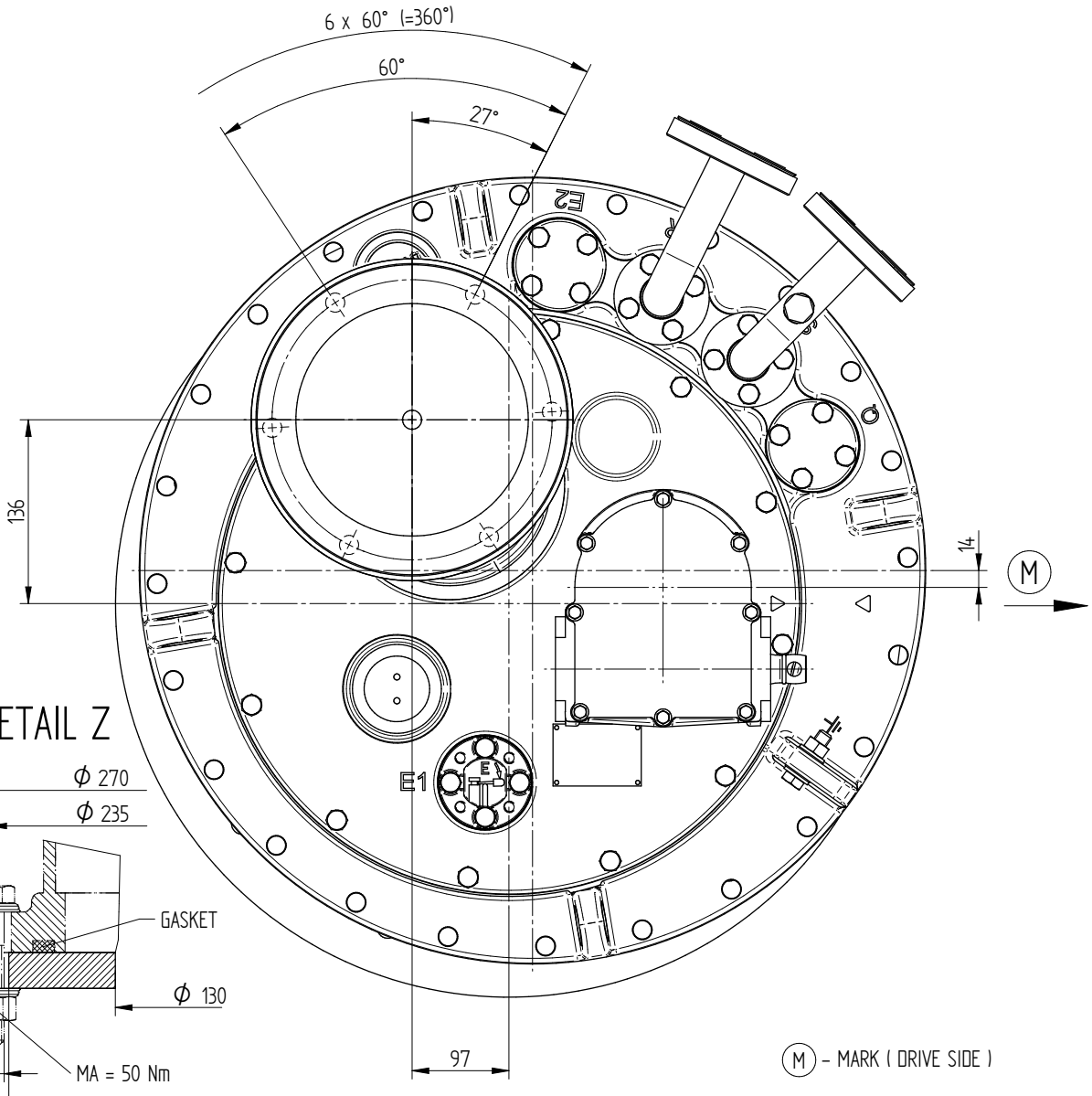
ON-LOAD TAP-CHANGER VACUTAP® VVS®
SWIVELLING RANGE FOR UPPER GEAR UNIT, SHAFT ON LEFT SIDE – HEAD DESIGN FOR SDS–

SERIAL NUMBER	
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MATERIAL NUMBER	SHEET
101168500E	1/1

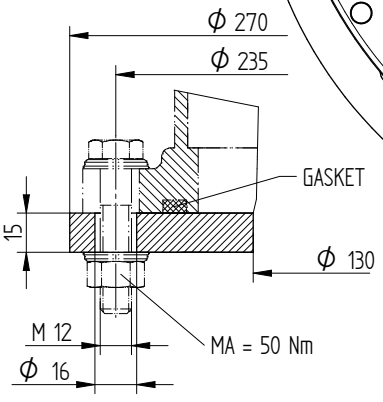
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DETAIL Z



(M) – MARK (DRIVE SIDE)

	DATE	NAME	DOCUMENT NO.	
DFTR.	06.06.2018	STREMPEL	SED 5995440 001 00	
CHKD.	06.06.2018	STREMPEL	CHANGE NO.	SCALE
STAND.	07.06.2018	PRODASTSCHUK	1087254	1:5

DIMENSION IN mm EXCEPT AS NOTED

ON-LOAD TAP-CHANGER VACUTAP® VVS®
PRV-FLANGE FOR PRESSURE RELIEF VALVE
MODEL WITH UPPER GEAR UNIT

SERIAL NUMBER
–

MATERIAL NUMBER
101168860E

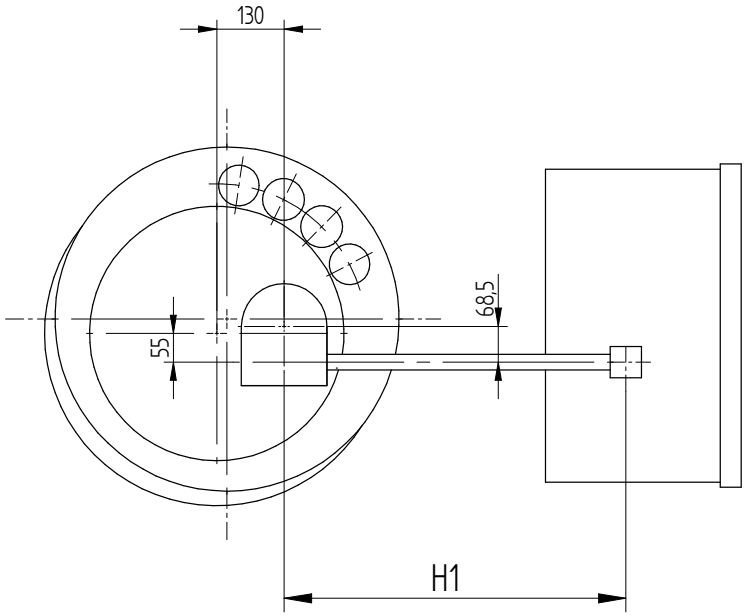
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1/1

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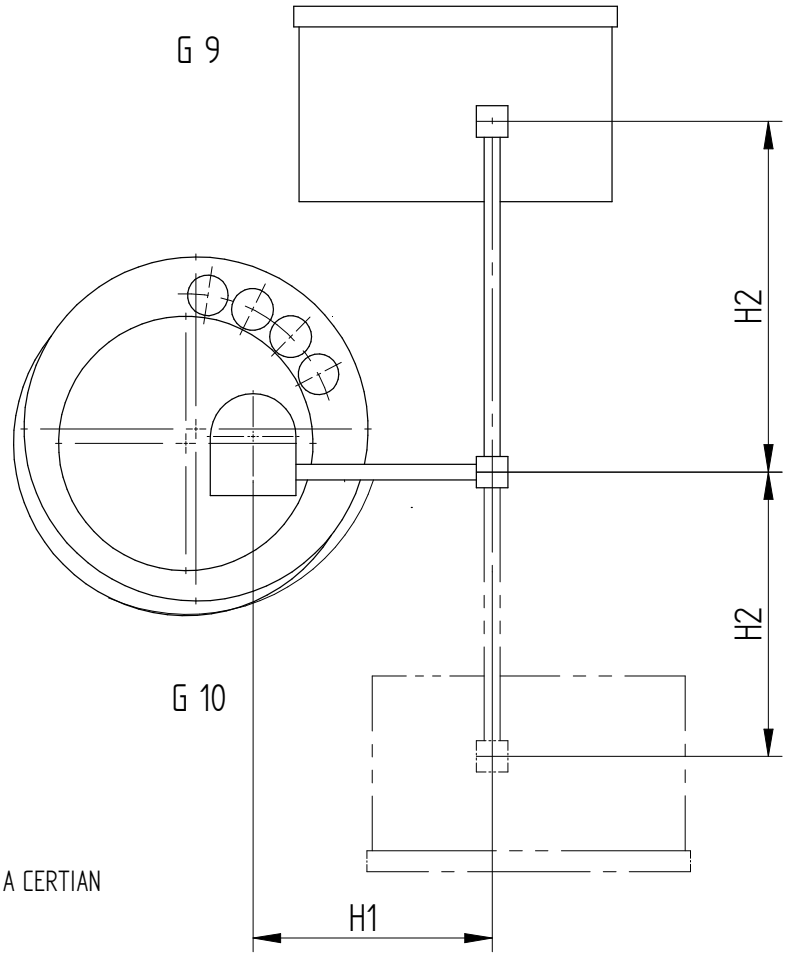
ARRANGEMENT G4

STANDARD DESIGN
(STRAIGHT DRIVE)



ARRANGEMENT G9, G10

(SPECIAL DESIGN)



ATTENTION:

H1 MAY BE REALIZED AS MINIMUM DIMENSIONS, POSSIBLY INCREASED BY A CERTIAN POSITION OF THE UPPER GEAR UNIT.

	MINIMUM DIMENSIONS		INTERMEDIATE BEARING REQUIRED FOR	
ARRANGEMENT	H1	H2	H1	H2
G4	≥ 400	-------	> 2254	-------
G9, G10	≥ 400	≥ 400	> 2309	2259

	DATE	NAME	DOCUMENT NO.	
DFTR.	17.10.2017	STEPHAN	SED 5705761 001 00	
CHKD.	17.10.2017	STREMPEL	CHANGE NO.	SCALE
STAND.	17.10.2017	PRODASTSCHUK	1084404	1:1

DIMENSION IN mm EXCEPT AS NOTED

ON-LOAD TAP-CHANGER VACUTAP® VVS®
HORIZONTAL DRIVING SHAFT
(STANDARD DESIGNS)

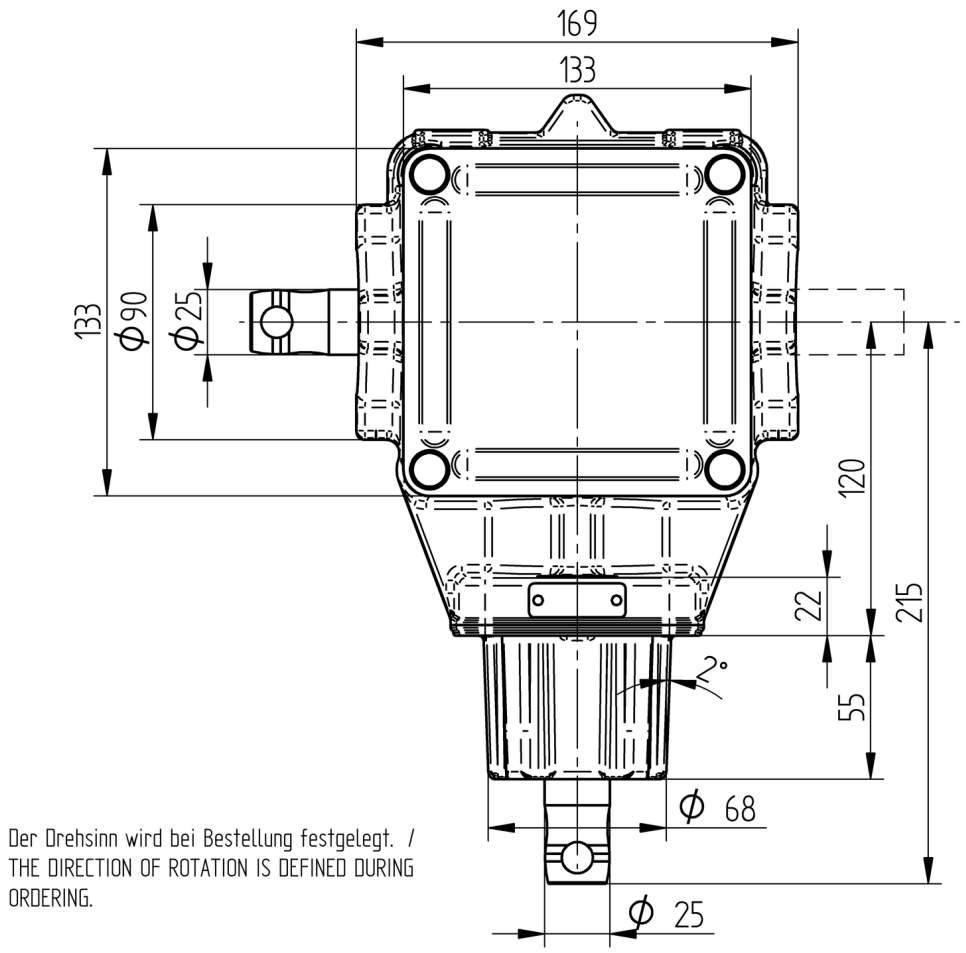
SERIAL NUMBER
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MATERIAL NUMBER
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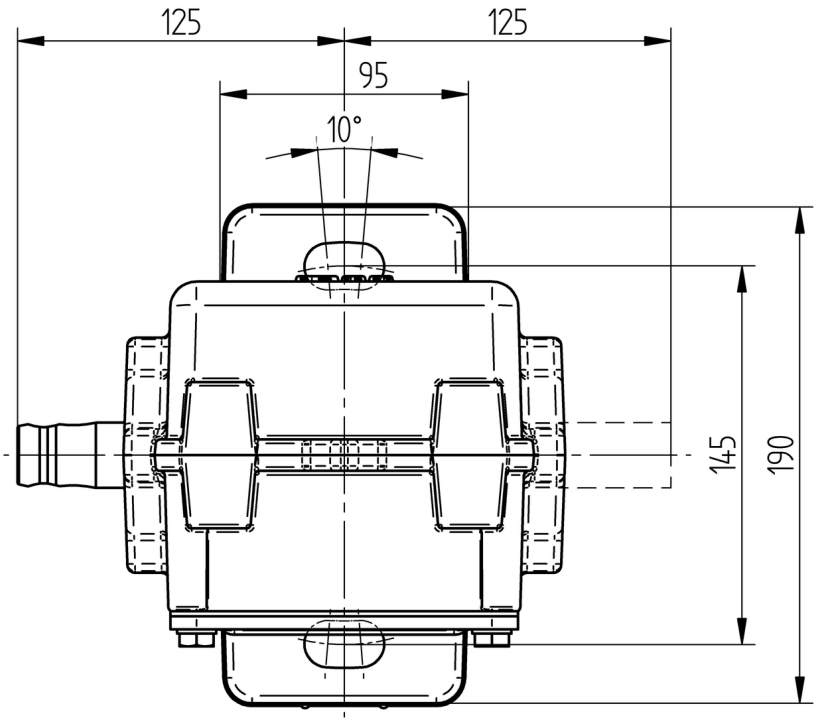
SHEET
1/1

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	Datum	Name	Dokumentnummer	
Gez.	01.06.2016	BRANDL	SED 2127250 000 02	
Gepr.	01.06.2016	HUBERTH	Änderungsnummer	Maßstab
Norm.	01.06.2016	PRODASTSCHUK	1074942	1:2

Maßangaben in mm, soweit nicht anders angegeben

Zubehör Stufenschalter
Winkeltrieb CD6400BEVEL GEAR CD6400
Maßzeichnung

Serialnummer
–

Materialnummer 8929167M

Blatt 1 / 1

A3

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